NEXTEER AUTOMOTIVE

Hot Zone CV Joint Production Line

Structured Project Document

Plant:	78
Product Family:	TRIPOT Hsg / CVOR (CV Joint Outer Race)
Process Area:	Hot Zone / Heat Treatment
Customers:	Nissan, Renault, Stellantis
Document Date:	15 October 2026
Source Material:	65 WhatsApp photographs of shop-floor documentation, HMI screens and equipment

This document consolidates every legible text block extracted from the photographic set: production work instructions, control plans, setup sheets, maintenance instructions, gage instructions, error-proofing procedures, safety fiches, OK/NOK visual standards, HMI readouts and machine identification plates. It also provides goals, features, technical specifications and a phased development road map for the line.

Nexteer Automotive — Hot Zone CV Joint Production Line

Structured Project Document (Compiled from 65 Site Photographs)

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1. PROJECT GOALS

G1	**Process Standardization**	Maintain and enforce controlled work instructions for wash, induction heat-treat, quench, and draw-oven operations across all CV joint part numbers.
G2	**Quality Control**	Implement multi-layer control plans (automatic PLC checks, manual operator checks, metrology gage checks) to guarantee part conformity.
G3	**Traceability**	Achieve full lot/part traceability via 2D DataMatrix codes, laser marking, and MES-style manual lot tracking.
G4	**Error-Proofing (EP)**	Reject non-conforming parts at geometry, 2D-code, and heat-treat-status levels before and after induction.
G5	**Safety & Compliance**	Enforce lockout/tagout, PPE, laser safety, and hot-part handling procedures per EHS standards.
G6	**Maintenance Excellence**	Execute 1st-level TPM routines (daily/weekly checks) on washers, induction machines, draw ovens, and conveyors.

2. PRODUCTION LINE — FEATURE OVERVIEW

2.1 Equipment Summary

Wash Station (TP & CV)	Part cleaning before heat treatment	OP_40010 / OP_400102; ALK-CAST 12.7 fluid; 54 ±3 °C; 27 s cycle; pressure max 3.5 bar / min 0.5 bar
Wash Blank (OR)	Blank wash prior to induction	OP_100102; CV OR #4; SV-000088
Induction Heat Treatment Machine (CV 1–CV 4)	Bell & Stem induction hardening	Siemens SINUMERIK (700826); dual-channel NC (BELL / STEM); Inducontrol HMI
Draw Oven (Tempering Furnace)	Continuous tempering after quench	OP_100402 Draw Oven CVOR #4; SV-000091; 3 zones @ 163 ±5 °C; Therelek Engineers make
Laser Marking Station	2D DataMatrix / part marking	172.38.22.39

2D Camera / Code Reader	DataMatrix verification	Keyence-style reader; AutoID Network Navigator (SR-20AH, IP 172.38.22.37)
Gage / OPD Check Station	Dimensional verification (OPD, spline)	Micrometer + pin gages; Go/NoGo spline gages
Unload / Boxing Station	Finished part boxing	Hot Zone reject bin + good-part boxing with 2D scan

2.2 Process Flow

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Blank Loading → Blank Wash (OP_100102)
    → 2D Code Scan / Lot Tracking
    → Induction Station: Bell Heat → Quench → Stem Heat → Quench (OP_100302/10030)
    → Draw Oven / Tempering (OP_100402)  [163 °C × 3 zones]
    → Cooling / Unloading
    → OPD & Spline Gage Check
    → Final Inspection (OCP) → Boxing (scan 2D code) → WIP Trolley
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2.3 Key Process Parameters (Bowl & Stem Recipe)

Highlighted (yellow) fields on the Setup Sheet are the **only parameters the operator may change**. All others require Engineering / ME intervention.

Z2 Axis Position	~123.5–144 mm (part-dep.)	~22–112.8 mm
Z Axis Scan Speed	9000 mm/min	9000 mm/min
Stop Time	4–6.5 ±0.5 s	4.3–5.6 ±0.5 s
Voltage %	65–85 ±5	75–85 ±5
Capacitor Selection	7 (part-dep.)	7 (part-dep.)
Bowl Frequency	15 ±1 kHz	—
Bell External Shower	100–105 LPM	—
Bell Middle Shower	35 ±5 LPM	—
Quench	Water 251, >5 bar zone Verte	Same

Exact Z2/Z3 positions to be verified with 1.5 mm Tlenlon gap check during setup.

2.4 Washing Parameters

Washing Temperature	54 ±3 °C
Washing Fluid	ALK-CAST 12.7
Chemical Concentration	2.8% (measured via conductivity meter / refractometer)
Drying Air Pressure	max 6 bar / min 4 bar
Fluid Pressure	max 3.5 bar / min 0.5 bar
Cycle Time	Correct 27 s

2.5 Draw Oven (Tempering) Parameters

Zone 1	163 ±5 °C	Axial blower 30 ±1.5 Hz
Zone 2	163 ±5 °C	Loading conveyor 50 ±2 Hz
Zone 3	163 ±5 °C	Furnace conveyor 42 ±2 Hz

3. QUALITY & CONTROL FEATURES

3.1 Control Plan (OCP34417178, Rev 06, 20/05/2025)

Automatic (100%) PLC Checks

1	Wash fluid pressure	max 3.5 / min 0.5 bar	100 %
2	Lot number	Correct PN scanned	Each box
3	Wash fluid temperature	54 ±3	100 %
4	Wash fluid level (tank)	Between Min/Max	100 %
5	Drying air nozzle pressure	max 6 / min 4 bar	100 %
6	Cycle time	Correct 27 s	100 %
7	Washer filter	Not clogged (machine alarm)	100 %

Manual Operator Checks (100 % Shift / Changeover)

8	Wash fluid pressure (visual)	max 3.5 / min 0.5 bar
9	Lot scanning (scanner)	Correct PN
10	Wash fluid temperature	54 ±3
11	Fluid level in totes	Min/Max
12	Drying air pressure	max 6 / min 4 bar
13	Chemical reference	ALK Cast 12.7
14	Fluid concentration (refractometer)	3.5–9.5 % (×2 for Aquaquench 251)
15	Part loading on conveyor	Present / Conforming / No damage
16	Part washed	No chips or oil
17	Part dried	No residual moisture
18	No residue / contaminants	Clean surface

Reaction Plan: Stop production, quarantine parts, do not ship, follow QA.01.00.00 "Non-conforming Material Management".

3.2 Error-Proofing (EP) Verification (EPR34409900, Rev 006, 01-07-2025)

Geometry EP

- Run master parts from table (page 2/2) and confirm they are **rejected**.

2D Code EP

1. Run a good part → **must be accepted**, PN correct.
2. Run a blank (no 2D code, laser disabled) → **must be rejected**.

3. Run master with damaged 2D code (laser disabled) → **must be rejected**.
4. Run already heat-treated part of same PN → **must be rejected**.

EP verification is performed **before the DataMatrix reader** at induction station entrance. On EP failure: inform Team Leader + call ME; do not start production until EP is positive.

3.3 OPD / Spline Gage Instructions

- Measure **OPD over 2 gage pins** on a lot of 5 parts before heat treatment and after heat treatment (same parts, 0° and 90° angles).
- Pin diameters per program table (e.g., 2.00–2.002 mm).
- Go gage full length → **Good part**; NoGo freely in full length → **Scrap / re-evaluate by QA, PE & ME**.
- If NoGo goes only 20 % max → keep as good part.
- Mark bearing face with a dot; parts numbered 1–5.

3.4 Gage SG 699436 (Max HUB Simulator)

1. Align gage spline with part spline; let part descend by own weight.
2. Check 8 mm thickness gage between thread end and gage surface → if thickness gage bigger than distance, **reject**.
3. Check with NoGo gage GA-780063 → gage must **not** go.
4. Mark OK part with "V" and place in WIP trolley.

3.5 OK vs NOK Visual Standards (VAS34402926_001, Zone: Hot Zone)

Inductor / coil alignment	Coil centered on part axis	Coil offset / tilted
Part seating on fixture	Part fully seated, centered	Part tilted, off-center
Bell quench shower contact	Uniform water pattern	Uneven / blocked nozzles
Heated part color (bell)	Uniform straw/gray	Spot overheating, scale, cracks
Stem heat zone	Clean heat band, no burn	Discoloration beyond band
2D code after mark	Readable, correct PN	Damaged, missing, wrong PN
Unloading orientation	Parts placed in big Hot Zone reject box / good box correctly	Parts dumped loosely

3.6 Crack Reaction Plan (GWI34417543)

If cracks appear: **stop machine, inform TL / QT / TE**, then:

1. Check quench concentration (manual refractometer)
2. Check quench drain due date
3. Check quench condition / contamination
4. Check paper filter presence & function
5. Check oil skimmer presence & function
6. Check shower flow, orientation, alignment, clogging
7. Check coil position, centering, alignment, condition
8. Check coil internal shower tube
9. Check process parameters
10. Check rings change follow-up
11. Check METLab results (case depth at 4 mm)

12. Check region S (heat, scratch, other surface defect)

13. Check MPI machine parameters (model, recipe, concentration, lighting)

14. Check blank chemical composition & blank part defects

Systemic actions: CRIB tool receiving inspection; coil condition control every last shift of the week; shower orientation/alignment control in PM2.

4. OPERATIONAL WORK INSTRUCTIONS (Documented)

STR34407290 / PRO.01.00.10	**Start-Stop Instruction**	Power on → check E-stops → pull air valve → HMI boot → Reset → Homing. Shutdown: push air valve, E-stop buttons, main power off.
RIC34402309 / RIC34401371	**Interrupted Cycle**	Manual mode → run pumps + all showers 3 min → open doors, remove parts (reuse non-HT, scrap HT parts) → Reset ×2 → restart. For simpler line: clear errors → Safety Reset → Auto-Start ×2.
GWI34413351	**Cleanup Instruction**	Start-of-shift general check; Friday deep clean exterior; cover all openings; clean polycarbonate screens with mild soap; no abrasives; no pressure jets near motors/electrical; avoid solvents.
CHO3440796-4	**Change Over Instruction**	Verify PN & mounted coil; change inductor, rings, bottom rings with correct Allen keys; disconnect/reconnect DM water & cooling; use teflon 1 mm gap for coil centering; load new recipe & verify Z2/Z3; run Learn on InduControl; fill Setup Check Sheet.
GWI3441351	**Sensibilité Check (TP & CV)**	Connect clips coil↔base / coil↔tail stock; close door, release E-stop, reset; verify "inductor touch" alarm on HMI for STEM and BOWL on both TP and CV. If NOK → inform TL, Maintenance & TE immediately.
VAS34402948	**Washer Nozzles Check**	Air-compress nozzles always horizontal; verify parts fully pass blow-off zone; if jammed, adjust nozzle angle so pieces pass smoothly; take before/after photos.
GWI34413698	**Hand Refractometer**	Run pumps 30 min for homogeneous concentration; zero the refractometer; sample from front-door port; for AQUAQUENCH 251 multiply reading ×2 for concentration %.
BLC34402308	**Traceability (Manual Lot)**	Directory Screen → TRACEABILITY → MANUAL LOT TRACKING → select PN → enter lot # twice (or scan label JUK) → submit.
—	**Non-Conforming Flow Chart**	Detect suspect product → identify & label → inform Team Leader / Technician Quality → re-verify → re-introduce only if conforming; else quarantine.

SOP34405838 / EHS.12.00.03	**Fiche de Sécurité (Workstation Safety)**	Risks: HMI impact, box-on-hold impact, AQUAQUENCH spill/slip, laser exposure, crushing/burns from moving/hot parts, noise >85 dB. PPE: gloves, safety shoes, anti-noise, safety glasses. Alert on safety-latch malfunction.

5. TECHNICAL SPECIFICATIONS

5.1 Network / IP Addressing (Machine Plate — EMMEPI)

NCU CP	192.168.214.1
MCP	192.168.214.192
IPC Ethernet 1	192.168.214.241
IPC Ethernet 2	10.0.0.98
NCU PN-IO	172.38.22.33
IM1	172.38.22.34
IM2	172.38.22.35
Converter CMS	172.38.22.36
2D Camera	172.38.22.37
Laser Sensor	172.38.22.38
Laser Marking Station	172.38.22.39
IO Link 1 / 2 / 3	172.38.22.40 / .41 / .42
DataMatrix Reader (AutoID)	172.38.22.37 (SR-20AH)

5.2 HMI / Control Systems

- **Siemens SINUMERIK** (labeled CV 4): Dual-channel NC programs (NC/MPF/BELL, NC/MPF/STEM). Active alarms observed: "BRAX % OUT OF RANGE: 3-3.5", "Read-in enable missing", "F12 dwell time".
- **Inducontrol** (SAET / IBI / SMITA): Process dashboard showing Voltage, Current, Power, Frequency, Bell Ext/Mid/Int Shower LPM, Stem Shower LPM, Quench Temp, Rotation RPM, Brix %, DM Conductivity; Job IDs for Bell & Stem; Heat Time 69.0 s (Bell) / 0.0 s (Stem); navigation to TRACEABILITY / OPERATIONS / RECIPE.
- **AutoID Network Navigator** (Keyence-style): Live view, auto focus, tuning banks 1–5, DataMatrix symbology, exposure 94–153 µs, HDR / Expand filters, matching-level graphs.

5.3 Safety / LOTO (Plant 78 — Continuous Draw Oven #5)

E-1	Main Electric	Turn off & lock electrical disconnect; verify pilot light off
A-1	Air (Compressor)	Turn off & lock safety valve; manually test air movements

2 locks required for complete controlled energy state. Notify affected personnel before servicing.

5.4 Parts & Programs Covered

Part programs include (non-exhaustive): CMP, CDV, XS2, FCA, DB35, XJI — sizes 23 st, 25 st, 26 st, 34UF, 66UF, 34GU. Customer PN ↔ Nexteer PN cross-reference maintained in Annex 1 (Customer to Nexteer PN table). EP master PN table (page 2/2 of EP instructions) links part length (±0.5 mm / ±0.25 mm) to master part numbers.

6. ROLES & RESPONSIBILITIES

Operator (OP)	Follow work instructions; change only yellow-marked setup parameters; perform 100 % manual checks; load/scan parts; react to alarms per instruction
Team Leader (TL)	First escalation point for NOK sensibility check, EP failure, interrupted cycle, non-conforming material
Maintenance (1st level)	Execute TPM sheets (TPM34411007, TPM34411499); check valves, E-stops, air pressure, water levels, filters, nozzles
Manufacturing Engineer (ME / TE)	Approve non-yellow parameter changes; correct EP failures; validate changeovers
Quality (QT / QA)	Control plan audits; gage/OPD correlation; non-conforming disposition per QA.01.00.00
Metrology	OPD & spline gage correlation; METLab case-depth results

7. DOCUMENT CONTROL INDEX

PRO.01.00.10	Start-Stop Instruction	3 — 25.10.2021
PRO.01.00.10	Washer Nozzles Check	15.01.2017
PRO.01.00.10	Cleanup Instruction	25.10.2019
PRO.01.00.10	Hand Refractometer	25.10.2020
PRO.01.00.10	Interrupted Cycle	25.10.2020
PRO.01.00.11	EP Verification Instructions	03 — 22.09.2021 / 006 — 01.07.2025
PRO.01.00.21	Spline Check Flow (Gage)	2 — 26.12.2019
PRO.01.00.07	Supplementary Instructions (Boxing)	05.02.2020
PRO.01.00.01	Operation Control Plan	3 — 05.03.2020 / 06 — 20.05.2025
GWI34417348	Set Up Sheet (Draw Oven)	05/12/2014 / 01
GWI34413351	Cleanup	23/05/2022 / 02
GWI3441351	Sensibilité Check	14/11/2015
GWI34413698	Hand Refractometer	6/7/2022 / 02
GWI34417543	Crack Reaction Plan	10/03/2025 / 5
CHO3440796	Change Over	02/2024
RIC34402309 / RIC34401371	Interrupted Cycle	2022 / 2024
STR34407290	Start-Stop	4/9/2022 / 02
BLC34402308	Manual Lot Traceability	12/11/2024 / 03
OCP34417178	Operation Control Plan	06 — 20/05/2025
EPR34409900	EP Verification	006 — 01-07-2025
GOP34413151 / GOP34413755	Gage Instructions	01 / 1

GOP34413754	Gage SG 699436	Rev 1 — 5/2/2025
SOP34405838 / EHS.12.00.03	Fiche de Sécurité	01 — 01.08.2024 / 11.03.2024
VAS34402926_001 / NPS.00.01.02	OK vs NOK Standards	21/02/2025 / 15.06.2022
VAS34402948	Washer Nozzles Check	16/2/2017
TPM34411007 / TPM34411499	1st Level Maintenance	03-01-2023
M.01.17.00	1st Level Maintenance (alt.)	Rev 1 — 03-01-2023

8. DEVELOPMENT / IMPROVEMENT ROADMAP

Framed as continuous-improvement actions for the Hot Zone line based on gaps observed in the photographed documentation set.

Phase 1 — Foundation & Digitization (0–3 months)

1.1	Digitize all paper "Controlled Copy" work instructions into a controlled EDMS (e.g., SharePoint / MES doc module) with version control and e-signature	Doc Control / IT	100 % of PRO/GWI/CHO/RIC docs searchable & revision-controlled
1.2	Build master data table: Customer PN ↔ Nexteer PN ↔ Recipe ↔ EP Master ↔ Gage pin (Annex 1 + EP table) as a single database	ME / IT	Single source of truth; no duplicate spreadsheets
1.3	Scan & index all 2D DataMatrix read results to MES; replace manual JUK lot entry where possible	IT / Controls	Auto lot tracking on ≥ 80 % of boxes
1.4	Standardize OK/NOK photo standards (VAS34402926) into digital tablets at each station	Quality	Real-time visual aid; photo log per shift

Phase 2 — Process Capability & TPM (3–6 months)

2.1	SPC on wash temp (54 ±3 °C), quench temp, concentration (3.5–9.5 %), shower LPM	Quality / ME	Cpk ≥ 1.33 on critical params
2.2	Predictive maintenance on induction coils (condition monitoring, ring change tracking)	Maintenance	Zero unplanned coil failures / quarter
2.3	Automated NoGo spline gage or in-line OPD sensor to replace manual pin check on high runners	ME / Capital	Gauge R&R < 10 %; cycle time neutral
2.4	Verify AutoID SR-20AH read rates ≥ 99.9 %; tune banks per part family	Controls	Read-rate dashboard; alarms < 0.1 %

Phase 3 — Error-Proofing & Traceability Upgrade (6–12 months)

3.1	Integrate EP verification routine into machine cycle (auto-test on changeover)	Controls / ME	EP documented automatically in MES
3.2	Link laser mark PN + heat-recipe-hash into 2D code; verify post-oven	IT / Quality	100 % post-oven code audit
3.3	Camera-based part-orientation check at unload (aligned with OK/NOK standards)	Controls	Eliminate manual orientation defects
3.4	Real-time OEE dashboard from SINUMERIK + Inducontrol tags	IT	OEE visible per shift; downtime coded

Phase 4 — Safety, EHS & Sustainability (12–18 months)

4.1	Digital LOTO checklist with photo capture of locked disconnects	EHS	100 % LOTO compliance audit score
4.2	Noise mapping & enclosure upgrades (docs cite >85 dB exposure)	EHS / Facilities	Measured < 85 dB at operator position
4.3	AQUAQUENCH spill containment & automated level alarms	Maintenance	Zero slip incidents; auto low-level alarm
4.4	Energy monitoring on draw oven (3 zones) & induction power factor	Facilities	≥ 5 % energy reduction vs. baseline

Phase 5 — Knowledge & Training (Ongoing)

5.1	Bilingual (FR/EN) operator competency matrix per document ID	Training / TL	Each operator signed off on all applicable docs
5.2	AR-assisted changeover guide (overlay on photos from CHO3440796)	ME / IT	Changeover time −20 %
5.3	Quarterly document review cycle; retire obsolete revisions	Doc Control	Zero expired "Controlled Copy" in circulation

9. RISK REGISTER (Top Items)

Incorrect coil / recipe after changeover	Non-uniform hardness, cracks	Changeover instruction + EP verification + master gage
Quench concentration drift	Soft spots / cracks	Refractometer 2×/week + PLC alarm + crack reaction plan
2D code unreadable at unload	Lost traceability, shipping hold	Dual reader (mark station + unload), NoGo code EP

Operator changes non-highlighted parameter	Out-of-spec heat treatment	HMI access levels; yellow-only edit policy; audit trail
Hot-part handling injury	Safety incident	Mandatory gloves (EHS fiche), interlocked doors, E-stop
Obsolete paper instructions on line	Wrong setup	Phase 1 digitization + quarterly review

10. APPENDIX — Image-to-Topic Map (All 65 Files)

4.20.41 PM	Wash blanks station CV OR #4, conveyor with parts, barcode scanner, red light curtain
4.20.42 PM (1)	AutoID Network Navigator HMI — DataMatrix live view, tuning banks, SR-20AH reader
4.20.42 PM (2)	Inside induction machine — pneumatics, blue hoses, red cover, E-stop
4.20.42 PM (3)	Induction station — gold cylinders, quench spray, green/blue/orange hoses
4.20.42 PM (4)	Induction bowl station close-up — oil/coolant, fixtures, hoses
4.20.42 PM	Stainless washer enclosure with viewing windows, yellow nozzles inside
4.20.43 PM (1)	Siemens SINUMERIK panel CV 4 — full keypad, E-stop, green pilot lights
4.20.43 PM (2)	Nexteer Production Work Instruction — SETUP SHEET bowl/stem recipes (yellow = operator-editable)
4.20.43 PM	Induction station — stem quench, ball joints visible, wet environment
4.20.44 PM (1)	Through wire-mesh guard: induction coil, cables, yellow frame
4.20.44 PM (2)	Draw Oven CVOR #4 control cabinets — 3 zone controllers, energy meter 354368, Therelek
4.20.44 PM (3)	Therelek draw oven exit — metal belt, ball studs on tray, gage instructions posted
4.20.44 PM (4)	HMI panel — manual/auto, homing, system reset, clear faults, E-stop, IP address list
4.20.44 PM	SINUMERIK screen: CHAN2/CHAN3 active, Z2/Z3 positions, NC blocks (STOPRE, G4 stop time, quench GOTO)
4.20.45 PM (1)	Nexteer Start-Stop Instruction (STR34407290) — photos of switches, HMI steps
4.20.45 PM	1st Level Maintenance Instruction (TPM34411007) — 14 steps, ALK-CAST 12.7, nozzles check
4.20.46 PM (1)	Washer Nozzles Check (VAS34402948) — orientation photos, before/after
4.20.46 PM	Washer Setup Sheet — washing temp, ALK-CAST 12.7, 2.8 %, conductivity meter
4.20.47 PM (1)	Operation Control Plan page 1 (OCP34417178) — automatic checks 1–7
4.20.47 PM	Instruction de Nettoyage — 10-step cleaning procedure for machine
4.21.00 PM (1)	Gage Instructions (GOP34413151) — OPD 0°/90°, pin gage table, micrometer photos
4.21.00 PM (2)	Instruction de Travail CVOR — load part, OCP checks, boxing steps (landscape)
4.21.00 PM (3)	Interrupted Cycle (BLC-style) — reset, homing, wash entry, lamp start (upside-down photo)
4.21.00 PM (4)	Operation Control Plan page 2 — manual checks 8–15
4.21.00 PM (5)	Operation Control Plan page 3 — checks 16–18 (washed, dried, no residue)
4.21.00 PM (6)	Annex 1 — Customer to Nexteer PN cross-reference (large table)
4.21.00 PM (7)	Annex 2 — Gage / fixture table (GA-780xxx pin numbers)
4.21.00 PM (8)	Instruction de Travail CVOR — full page with EP & OCP steps (landscape)
4.21.00 PM (9)	Production Work Instruction — Work Place Organization / 5S checklist (landscape)
4.21.00 PM	Supplementary Instructions — Manual Lot Tracking screenshots (upside-down)
4.21.01 PM (1)	Cleanup Instruction (GWI34413351) — photos of induction machine interior

4.21.01 PM (10)	1st Level Maintenance (alt. SV 000006/114/115/90) — 17 steps, quench water, foam check
4.21.01 PM (2)	Sensibilité Check Instruction TP & CV (GWI3441351) — clip connections, inductor touch alarm
4.21.01 PM (3)	Change Over Instruction (CHO3440796-4) page 13 — coil, rings, bottom rings, Allen keys
4.21.01 PM (4)	Change Over page 23 — master placement, teflon gap, Z2/Z3, Bell/STEM recipes
4.21.01 PM (5)	Change Over page 23 (cont.) — Learn on Inducontrol, groups, single cycle, setup sheet
4.21.01 PM (6)	EP Verification Instructions (EPR34409900) page 1/2 — geometry & 2D code error proofing
4.21.01 PM (7)	EP Verification page 2/2 — master part numbers by program (CDV, CMP, XS2, FCA, DB35)
4.21.01 PM (8)	Hand Refractometer Instructions (GWI34413698) — sampling, ×2 Aquaquench 251
4.21.01 PM (9)	Supplementary / Crack Reaction Plan (GWI34417543) — 14 checks + systemic actions
4.21.01 PM	Supplementary Instructions — boxing orientation GOOD/BAD (red/green boxes, upside-down)
4.21.02 PM (1)	Interrupted Cycle (RIC34402309) — manual mode, 3-min showers, hot-part glove warning
4.21.02 PM (10)	IP address plate (EMMEPI) — full network table
4.21.02 PM (2)	OK vs NOK — inductor coil alignment
4.21.02 PM (3)	Interrupted Cycle (RIC34401371) — clear errors, Safety Reset, Auto-Start ×2
4.21.02 PM (4)	Safety LOTO placard — Draw Oven #5, E-1 + A-1, 2 locks required
4.21.02 PM (5)	OK vs NOK — part seating on fixture (straight vs tilted)
4.21.02 PM (6)	OK vs NOK — bell quench surface condition
4.21.02 PM (7)	OK vs NOK — heated part / scale defects
4.21.02 PM (8)	Fiche de Sécurité au Poste de Travail (SOP34405838) — risks & PPE
4.21.02 PM (9)	Fiche de Sécurité (alt. orientation) — same risk matrix
4.21.02 PM	Inducontrol HMI — shower flow graph, Rejected status, threshold 300 L/min
4.21.03 PM (1)	Non-conforming material handling flow chart (landscape, blurry)
4.21.03 PM (10)	Fiche de Sécurité (rotated) — same content
4.21.03 PM (2)	Gage Instructions + Non-conforming flow (two pages in sleeve)
4.21.03 PM (3)	Non-conforming material handling flow chart (clearer)
4.21.03 PM (4)	1st Level Maintenance (TPM34411499) — 8 steps, air valve, E-stop, 3 fans, 6-pcs belt check
4.21.03 PM (5)	Instruction de Nettoyage (rotated, in sleeve)
4.21.03 PM (6)	Gage Instructions — Spline Check Flow for CV OR (GOP34413755) — decision flowchart
4.21.03 PM (7)	Gage Instructions SG 699436 (GOP34413754) — Max HUB simulator, 4-step check
4.21.03 PM (8)	Inducontrol process dashboard — V/I/P/F, showers, quench 34.6 °C, RPM 112, Brix 3.7 %, conductivity 200, Job IDs, heat time 69 s
4.21.03 PM (9)	Set Up Sheet (GWI34417348) — draw oven 163 ±5 °C ×3 zones, blower/conveyor Hz
4.21.03 PM	Nexteer Automotive main HMI — Traceability / Operations / Recipe buttons, 3D line model

End of document. Generated from visual content of 65 photographs. Values transcribed as observed; verify against master controlled copies before operational use.

APPENDIX B - FULL PHOTOGRAPHIC GALLERY

All 64 source photographs reproduced below at fitted size, each with its filename and the content extracted from it (same order as the Appendix A map table). Images are auto-rotated using EXIF data where available.

[1/64] **WhatsApp Image 2026-10-15 at 4.20.41 PM.jpeg**

Content: Wash blanks station CV OR #4, conveyor with parts, barcode scanner, red light curtain



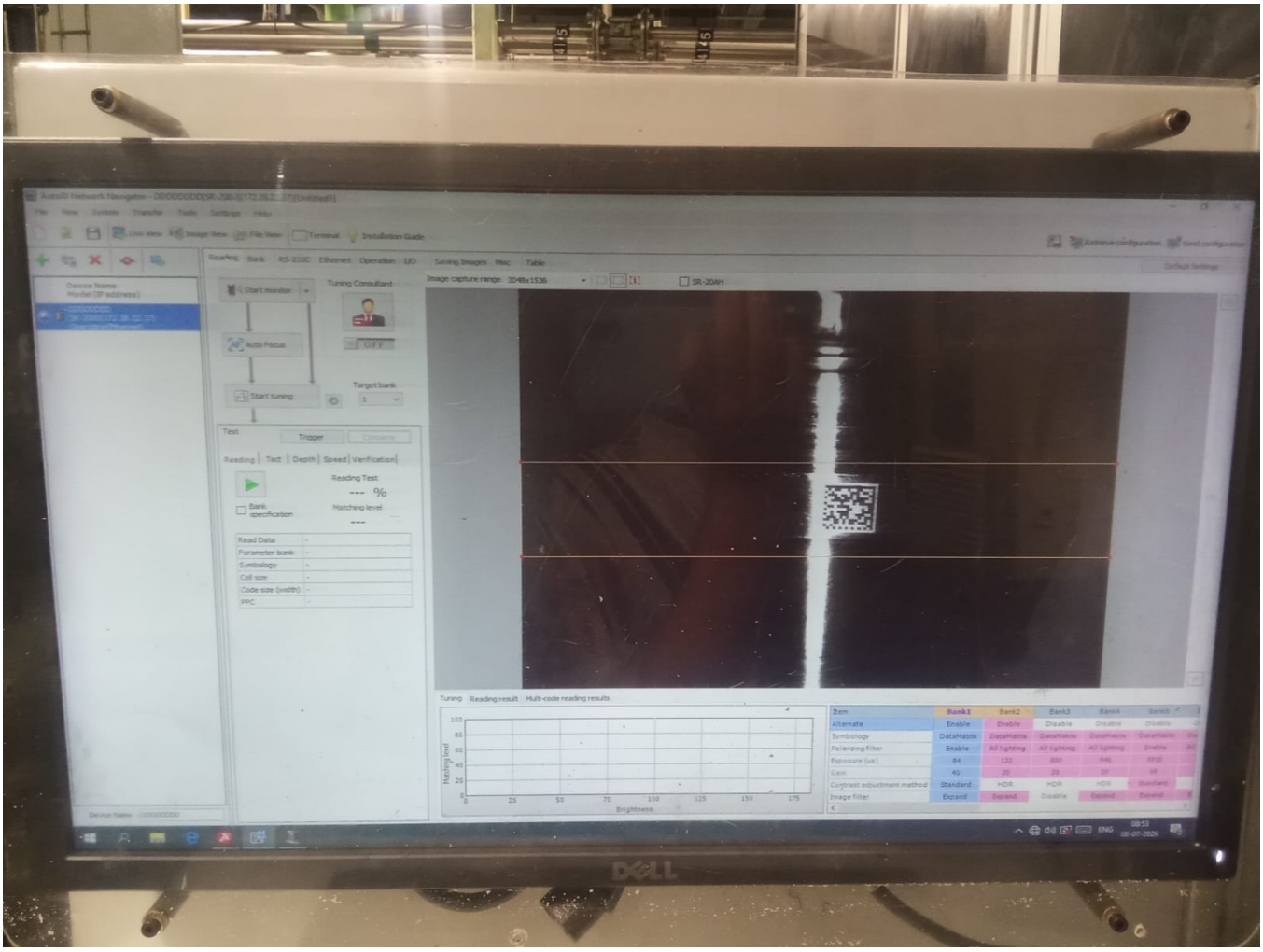
[2/64]**WhatsApp Image 2026-10-15 at 4.20.42 PM.jpeg**

Content: Stainless washer enclosure with viewing windows, yellow nozzles inside



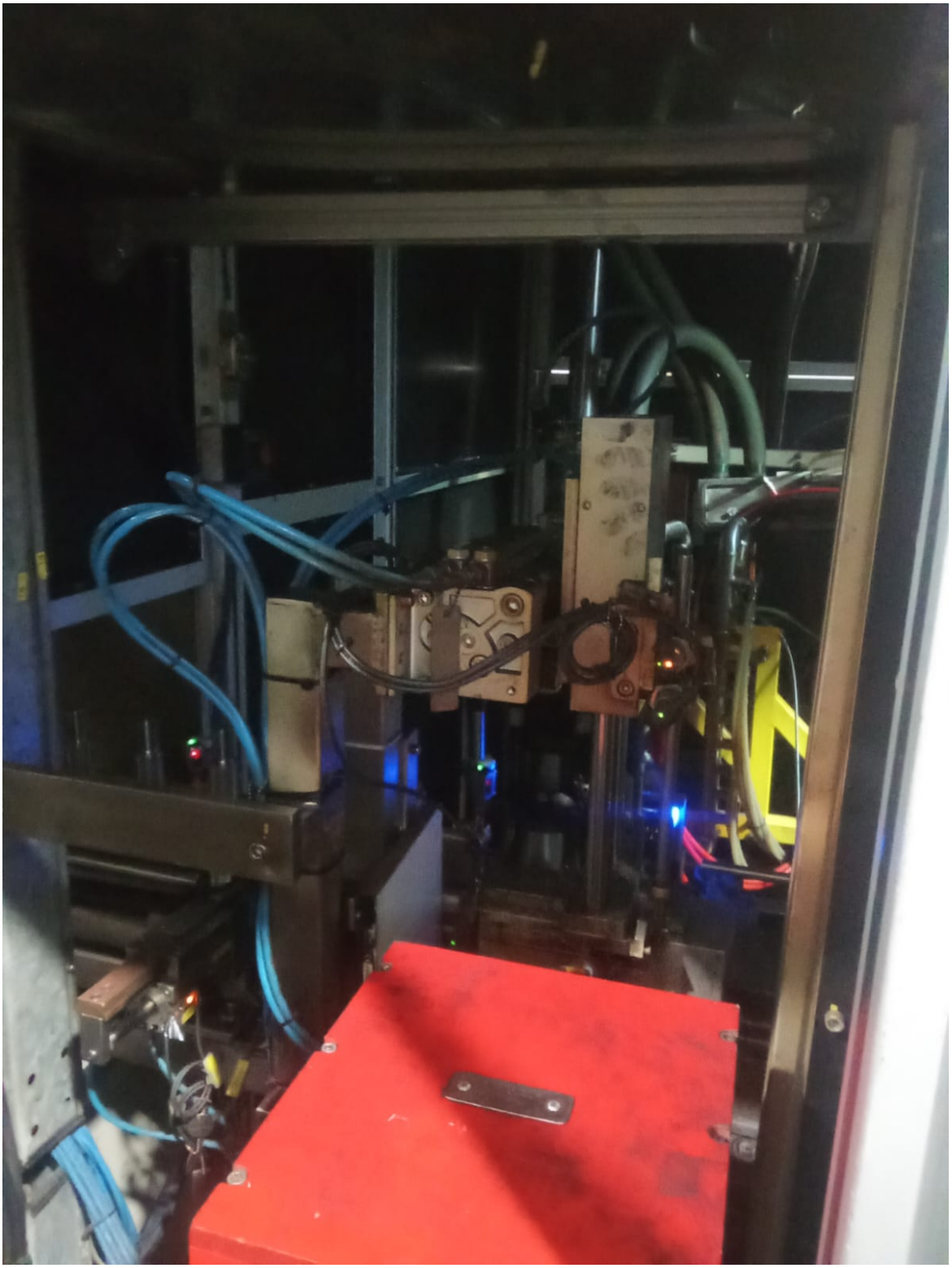
[3/64]**WhatsApp Image 2026-10-15 at 4.20.42 PM (1).jpeg**

Content: AutoID Network Navigator HMI — DataMatrix live view, tuning banks, SR-20AH reader



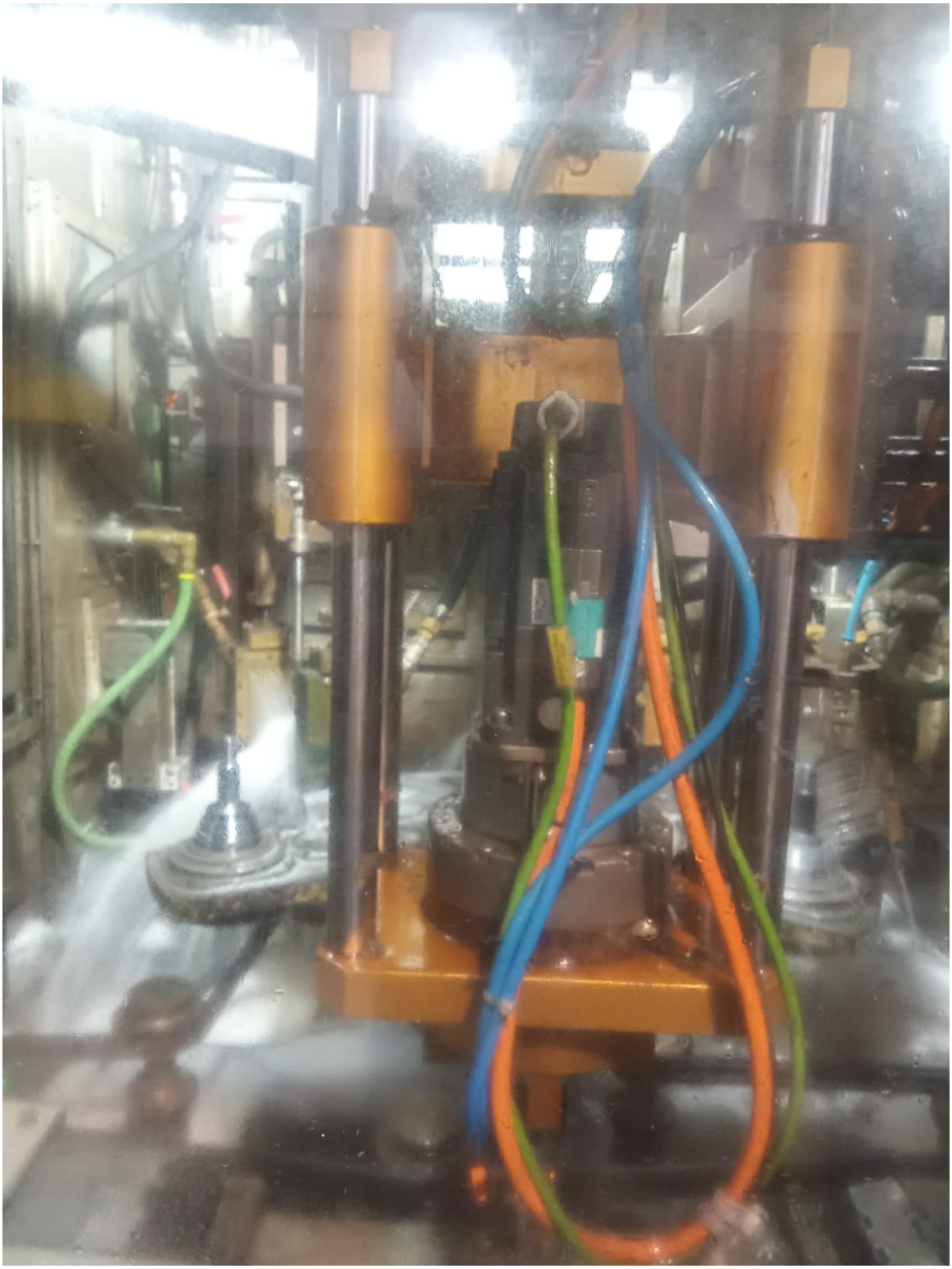
[4/64]**WhatsApp Image 2026-10-15 at 4.20.42 PM (2).jpeg**

Content: Inside induction machine — pneumatics, blue hoses, red cover, E-stop



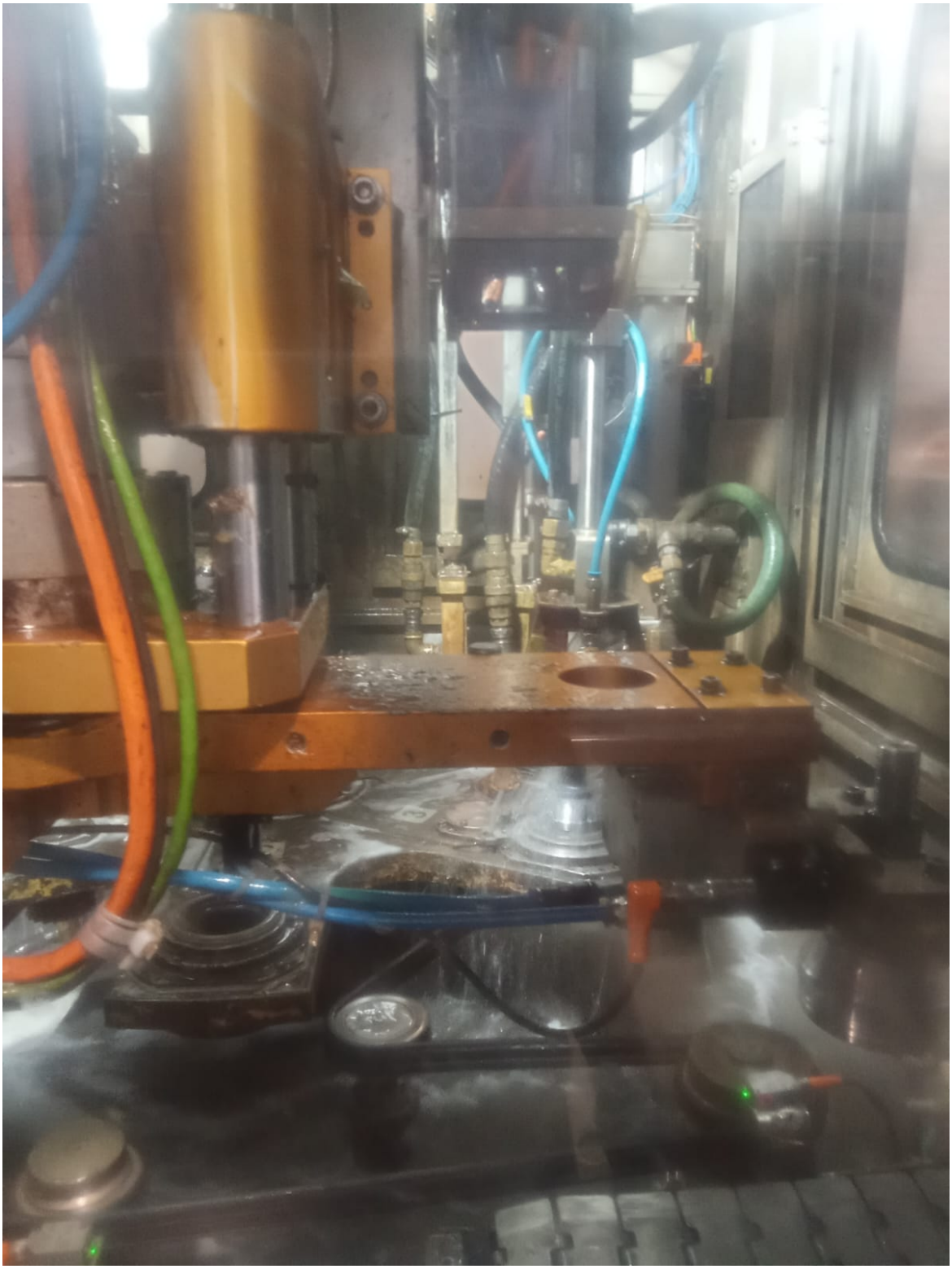
[5/64]**WhatsApp Image 2026-10-15 at 4.20.42 PM (3).jpeg**

Content: Induction station — gold cylinders, quench spray, green/blue/orange hoses



[6/64]**WhatsApp Image 2026-10-15 at 4.20.42 PM (4).jpeg**

Content: Induction bowl station close-up — oil/coolant, fixtures, hoses



[7/64]**WhatsApp Image 2026-10-15 at 4.20.43 PM.jpeg**

Content: Induction station — stem quench, ball joints visible, wet environment



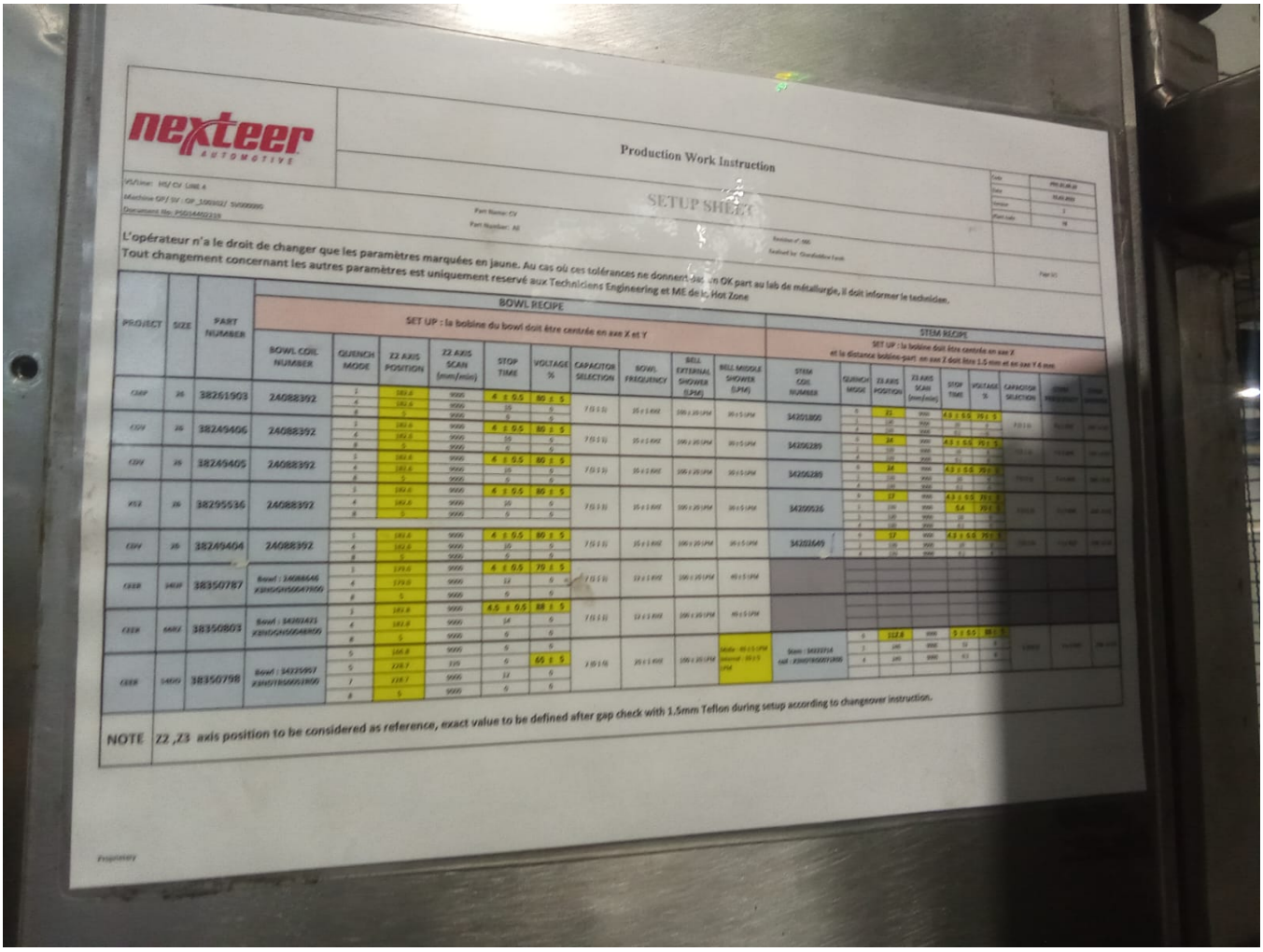
[8/64]**WhatsApp Image 2026-10-15 at 4.20.43 PM (1).jpeg**

Content: Siemens SINUMERIK panel CV 4 — full keypad, E-stop, green pilot lights



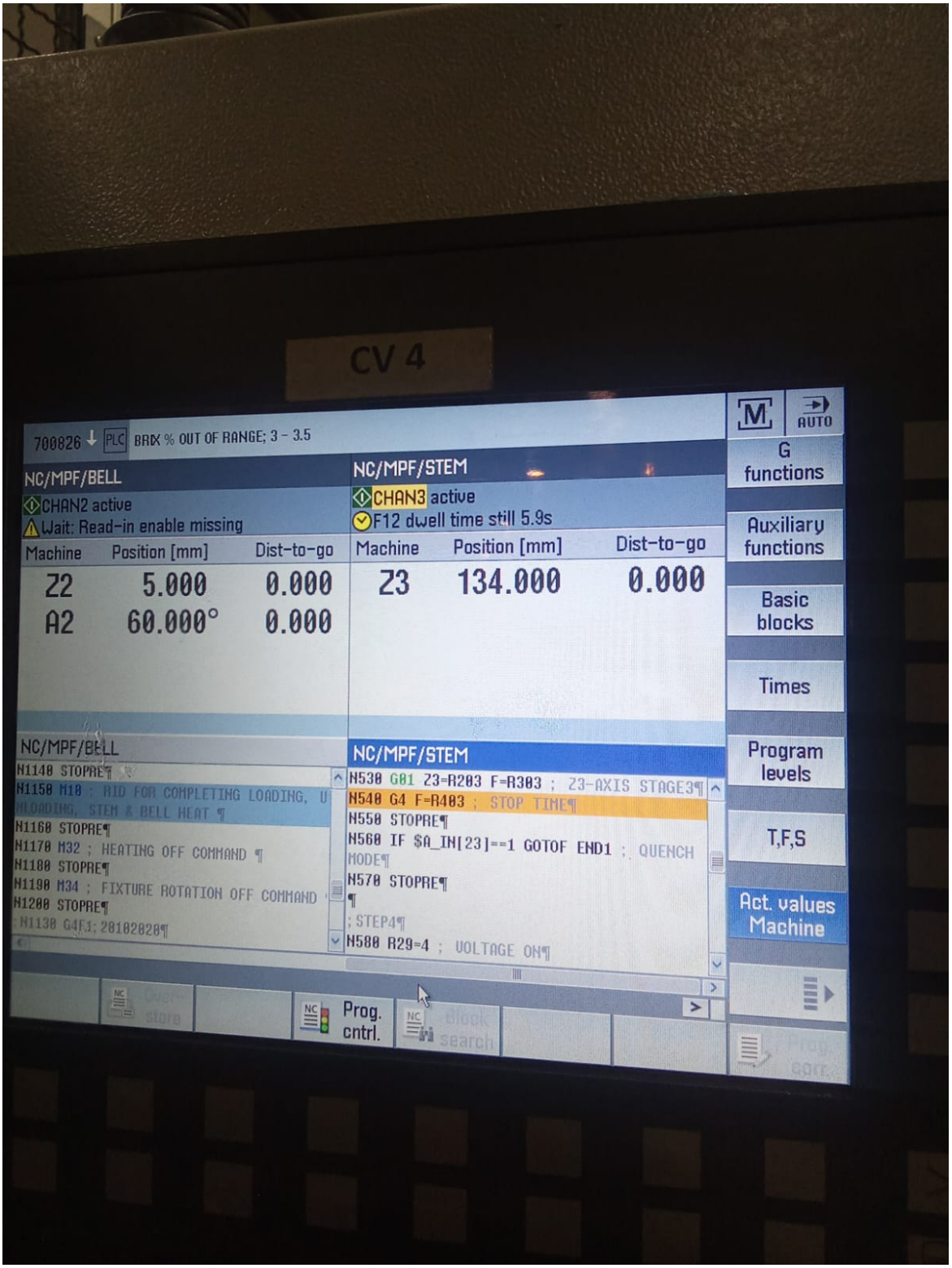
[9/64]**WhatsApp Image 2026-10-15 at 4.20.43 PM (2).jpeg**

Content: Nexteer Production Work Instruction — SETUP SHEET bowl/stem recipes (yellow = operator-editable)



[10/64] **WhatsApp Image 2026-10-15 at 4.20.44 PM.jpeg**

Content: SINUMERIK screen: CHAN2/CHAN3 active, Z2/Z3 positions, NC blocks (STOPRE, G4 stop time, quench GOTO)



[11/64] **WhatsApp Image 2026-10-15 at 4.20.44 PM (1).jpeg**

Content: Through wire-mesh guard: induction coil, cables, yellow frame



[12/64] **WhatsApp Image 2026-10-15 at 4.20.44 PM (2).jpeg**

Content: Draw Oven CVOR #4 control cabinets — 3 zone controllers, energy meter 354368, Therelek



[13/64] **WhatsApp Image 2026-10-15 at 4.20.44 PM (3).jpeg**

Content: Therelek draw oven exit — metal belt, ball studs on tray, gage instructions posted



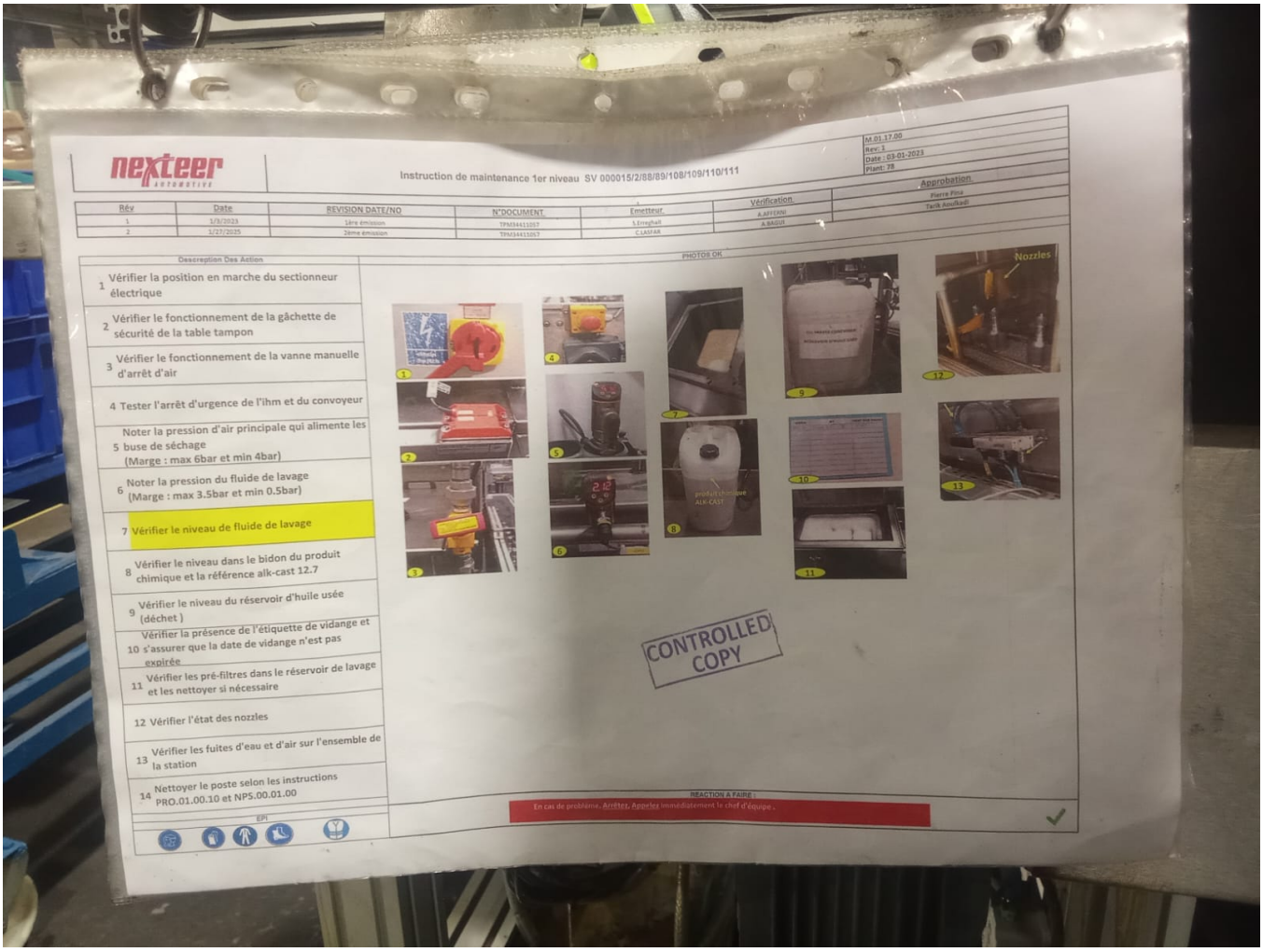
[14/64] **WhatsApp Image 2026-10-15 at 4.20.44 PM (4).jpeg**

Content: HMI panel — manual/auto, homing, system reset, clear faults, E-stop, IP address list



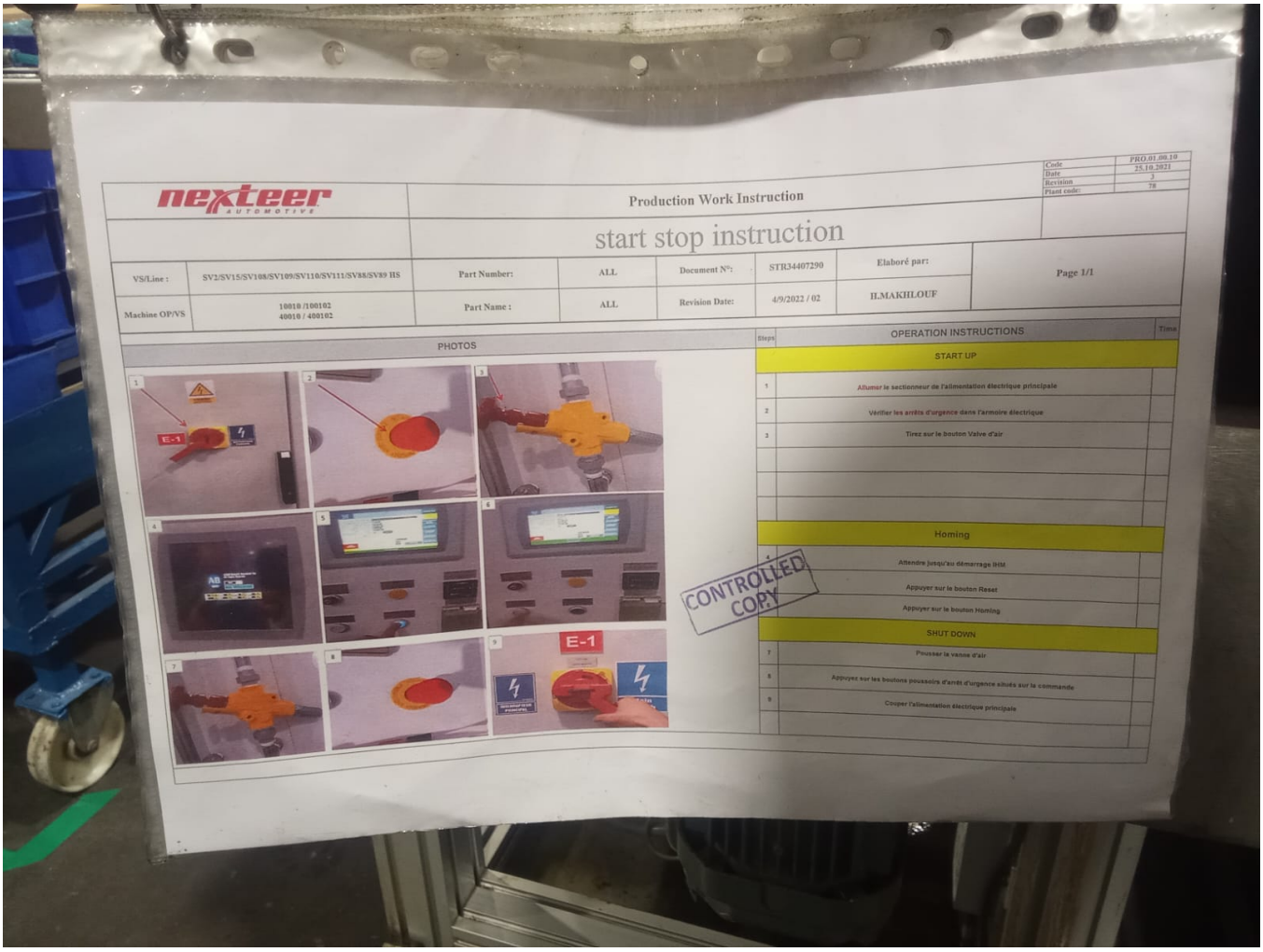
[15/64] **WhatsApp Image 2026-10-15 at 4.20.45 PM.jpeg**

Content: 1st Level Maintenance Instruction (TPM34411007) — 14 steps, ALK-CAST 12.7, nozzles check



[16/64] **WhatsApp Image 2026-10-15 at 4.20.45 PM (1).jpeg**

Content: Nexteer Start-Stop Instruction (STR34407290) — photos of switches, HMI steps



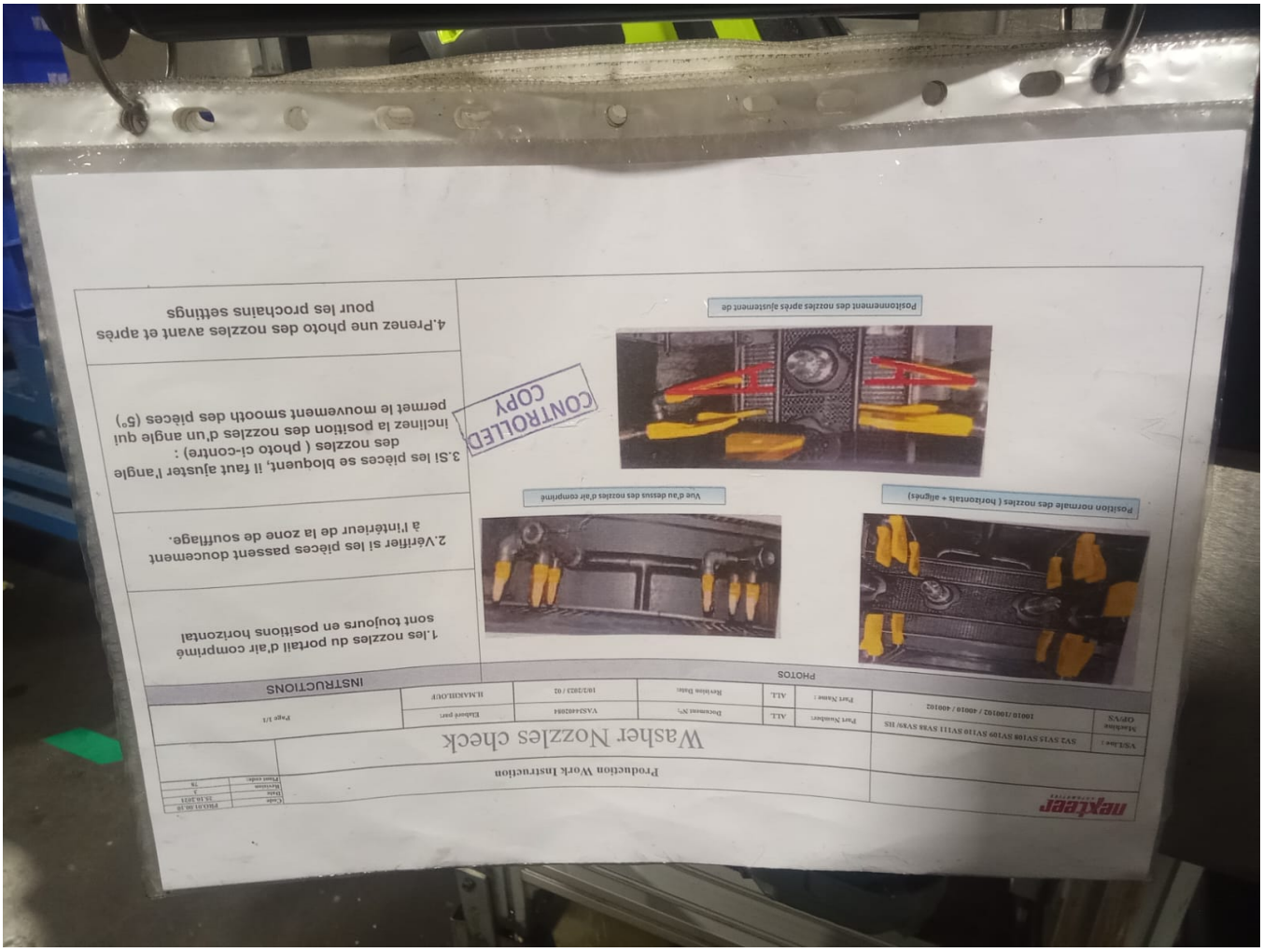
[17/64] **WhatsApp Image 2026-10-15 at 4.20.46 PM.jpeg**

Content: Washer Setup Sheet — washing temp, ALK-CAST 12.7, 2.8 %, conductivity meter



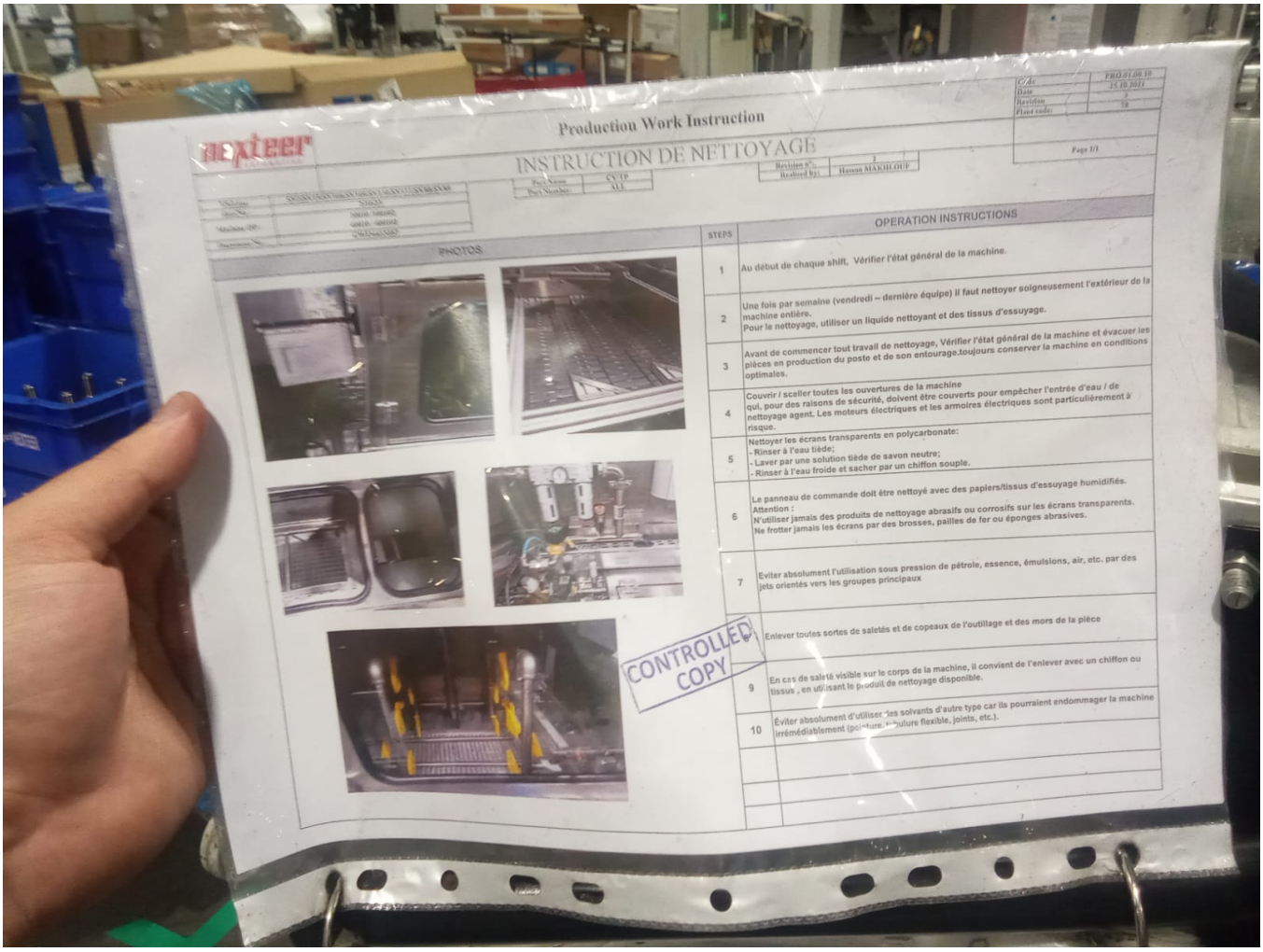
[18/64] **WhatsApp Image 2026-10-15 at 4.20.46 PM (1).jpeg**

Content: Washer Nozzles Check (VAS34402948) — orientation photos, before/after



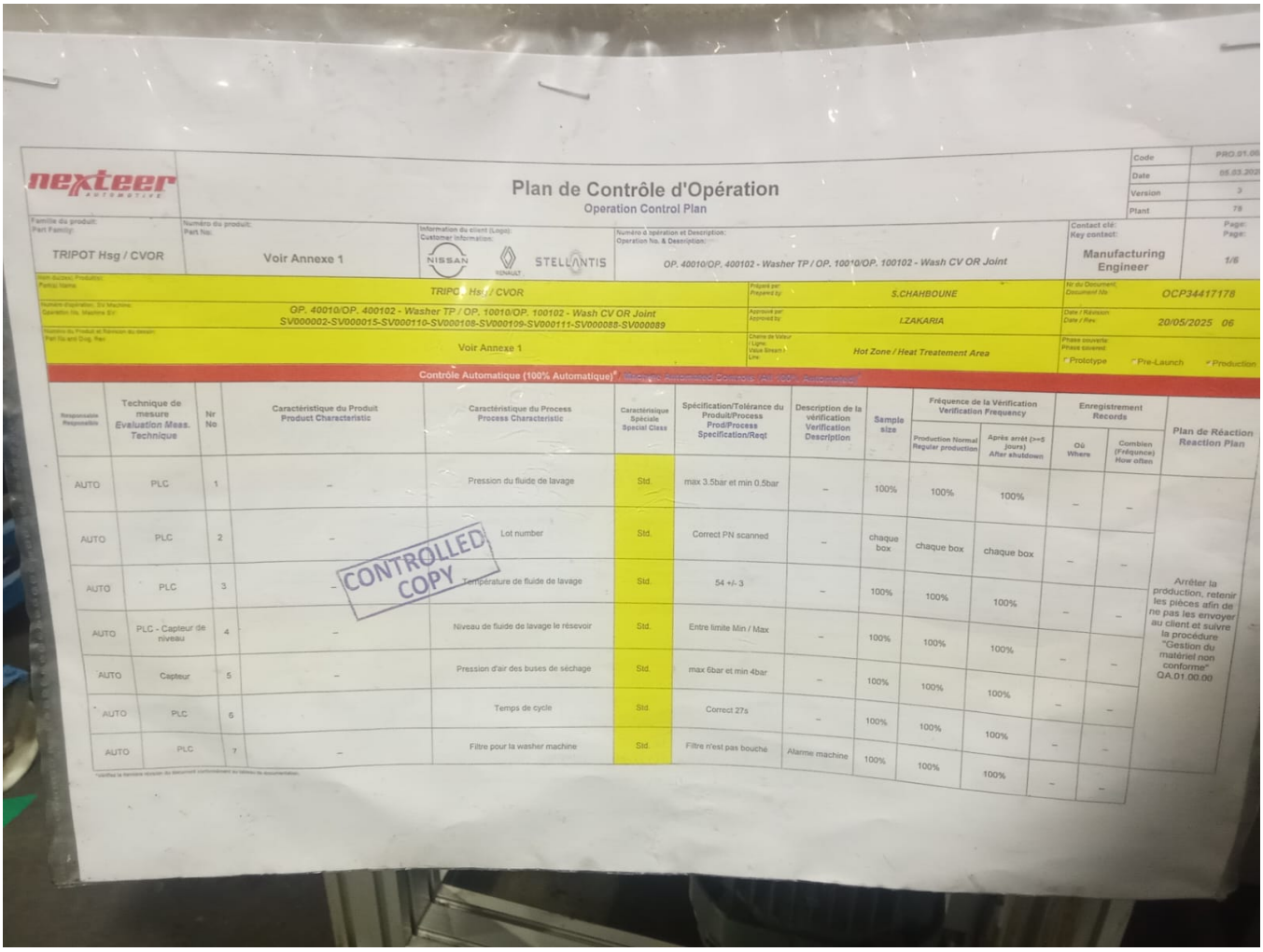
[19/64] **WhatsApp Image 2026-10-15 at 4.20.47 PM.jpeg**

Content: Instruction de Nettoyage — 10-step cleaning procedure for machine



[20/64] **WhatsApp Image 2026-10-15 at 4.20.47 PM (1).jpeg**

Content: Operation Control Plan page 1 (OCP34417178) — automatic checks 1–7



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Plan de Contrôle d'Opération
Operation Control Plan

Code	PRO.01.00
Date	05.03.2020
Version	3
Plant	78
Page:	1/6

Famille du produit / Part Family	Numéro du produit / Part No	Information du client (Logo) / Customer Information	Numéro d opération et Description / Operation No. & Description	Contact clé / Key contact
TRIPOT Hsg / CVOR	Voir Annexe 1	NISSAN · RENAULT · STELLANTIS	OP. 40010/OP. 400102 - Washer TP / OP. 100*0/OP. 100102 - Wash CV OR Joint	Manufacturing Engineer

Nom du(des) Produit(s) / Part(s) Name	TRIPOT Hsg / CVOR	Préparé par / Prepared by: S.CHAHBOUNE	Nr du Document / Document Nbr: OCP34417178
Numéro d opération, Du Machine / Operation No, Machine Nr	OP. 40010/OP. 400102 - Washer TP / OP. 10010/OP. 100102 - Wash CV OR Joint SV000002-SV000015-SV000110-SV000108-SV000109-SV000111-SV000088-SV000089	Approuvé par / Approved by: I.ZAKARIA	Date / Révision / Date / Rev. 20/05/2025 06
Numéro du Produit et Révision du dessin / Part No and Dwg. Rev	Voir Annexe 1	Chaine de Valeur / Ligne / Value Stream / Line: Hot Zone / Heat Treatement Area	Phase couverte / Phase covered: Prototype · Pre-Launch · Production

Contrôle Automatique (100% Automatique)* / Machine Automated Control, All 100% Automated*

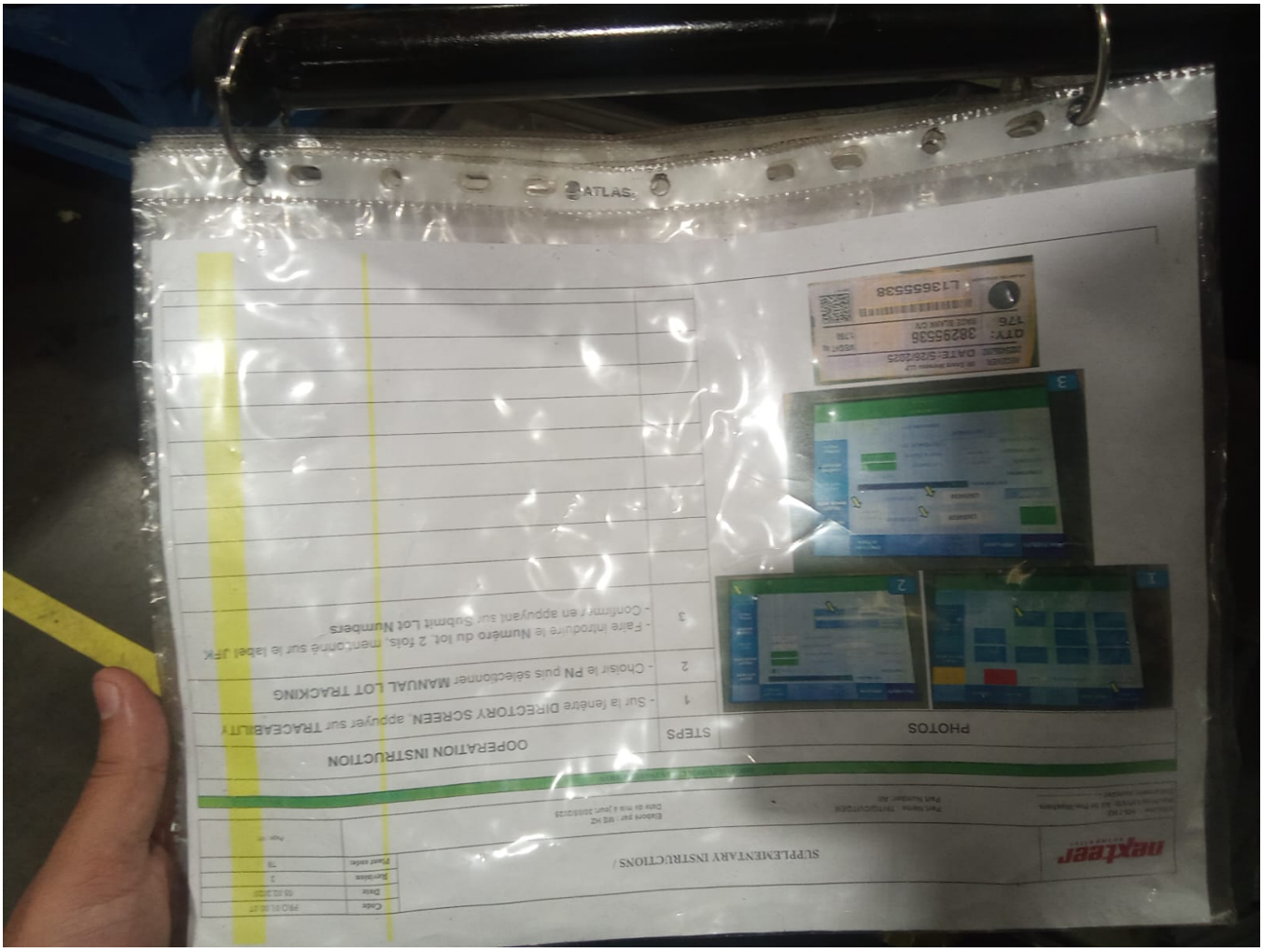
Responsable / Responsible	Technique de mesure / Evaluation Meas. Technique	Nr / No	Caractéristique du Produit / Product Characteristic	Caractéristique du Process / Process Characteristic	Caractéristique Spéciale / Special Class	Spécification/Tolérance du Produit/Process / Prod/Process Specification/Reqt	Description de la vérification / Verification Description	Sample size	Fréquence de la Vérification / Verification Frequency: Production Normal / Regular production	Fréquence de la Vérification / Verification Frequency: Après arrêt (>=5 jours) / After shutdown	Enregistrement / Records: Où / Where	Enregistrement / Records: Combien (Fréquence) / How often	Plan de Réaction / Reaction Plan
AUTO	PLC	1	–	Pression du fluide de lavage	Std.	max 3.5bar et min 0.5bar	–	100%	100%	100%	–	–	Arrêter la prôduction, retenir les pièces afin de ne pas les envoyer au client et suivre la procédure "Gestion du matériel non conforme" QA.01.00.00
AUTO	PLC	2	–	Lot number	Std.	Correct PN scanned	–	chaque box	chaque box	chaque box	–	–	
AUTO	PLC	3	–	Température de fluide de lavage	Std.	54 +/- 3	–	100%	100%	100%	–	–	
AUTO	PLC - Capteur de niveau	4	–	Niveau de fluide de lavage le résevoir	Std.	Entre limite Min / Max	–	100%	100%	100%	–	–	
AUTO	Capteur	5	–	Pression d'air des buses de séchage	Std.	max 6bar et min 4bar	–	100%	100%	100%	–	–	
AUTO	PLC	6		Temps de cycle	Std.	Correct 27s	–	100%	100%	100%	–	–	
AUTO	PLC	7	–	Filtre pour la washer machine	Std.	Filtre n'est pas bouché	Alarme machine	100%	100%	100%	–	–	

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*Vérifier la dernière révision du document conformément au tableau de documentation.

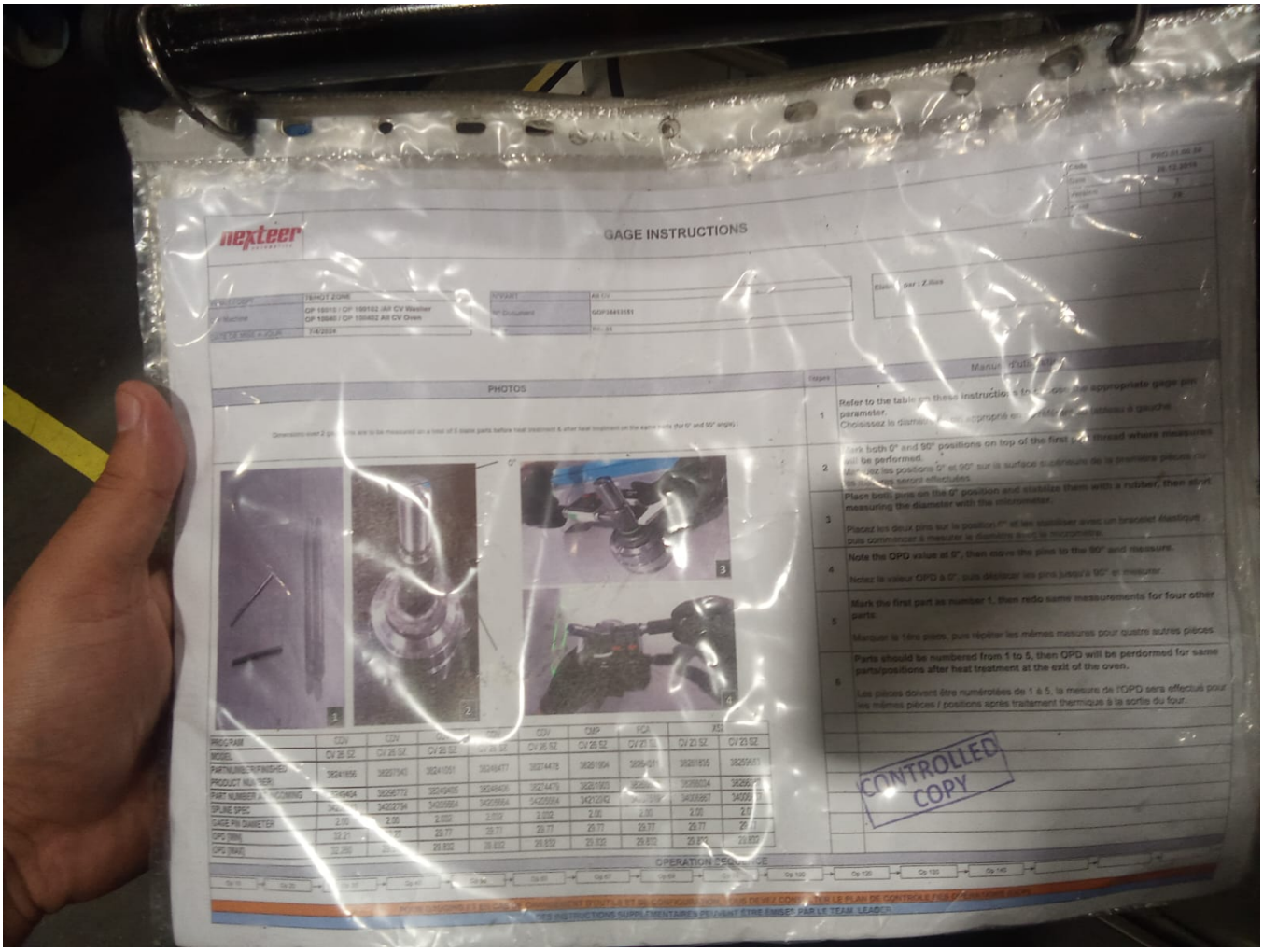
[21/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM.jpeg**

Content: Supplementary Instructions — Manual Lot Tracking screenshots (upside-down)



[22/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (1).jpeg**

Content: Gage Instructions (GOP34413151) — OPD 0°/90°, pin gage table, micrometer photos



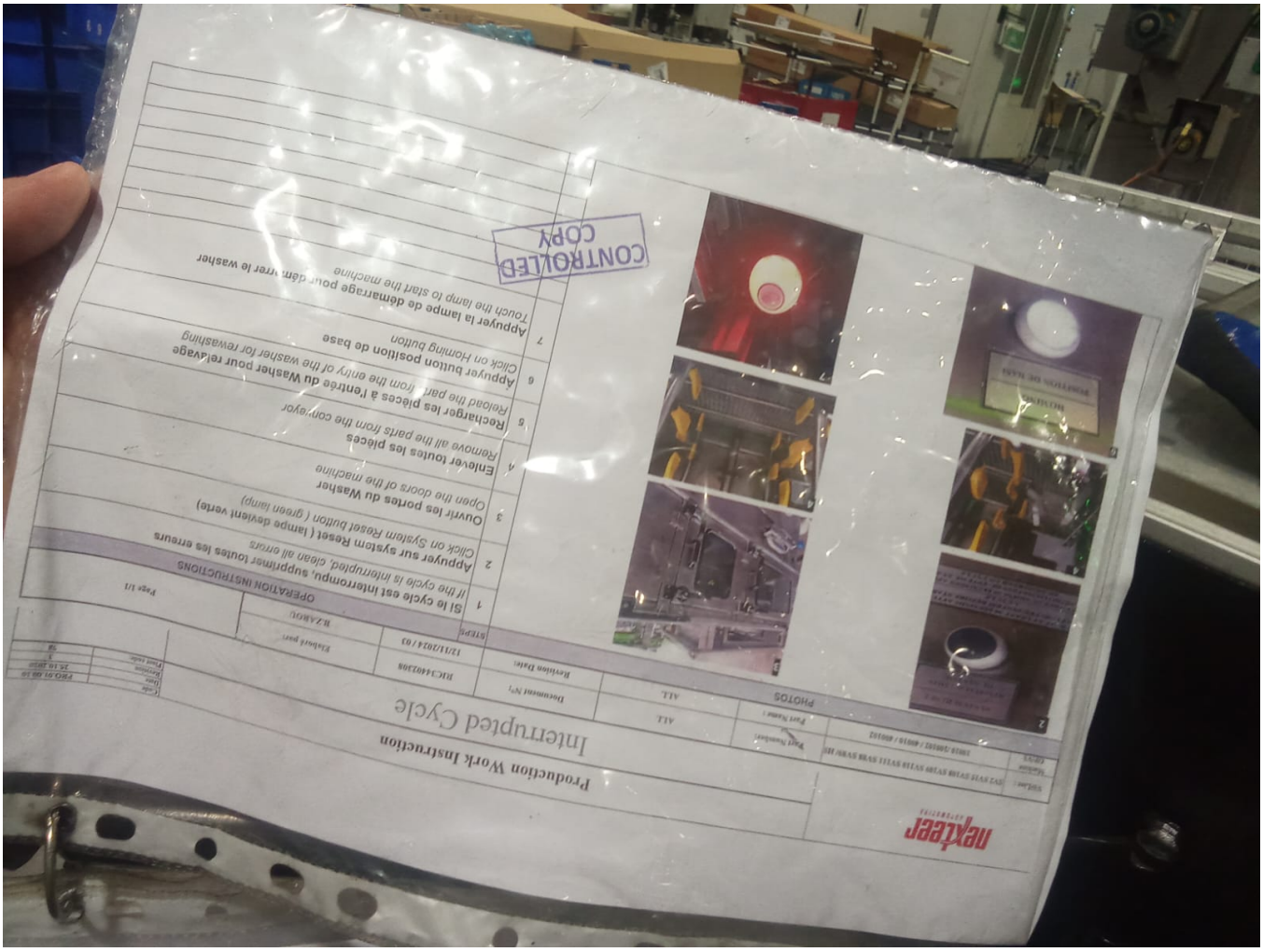
[23/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (2).jpeg**

Content: Instruction de Travail CVOR — load part, OCP checks, boxing steps (landscape)



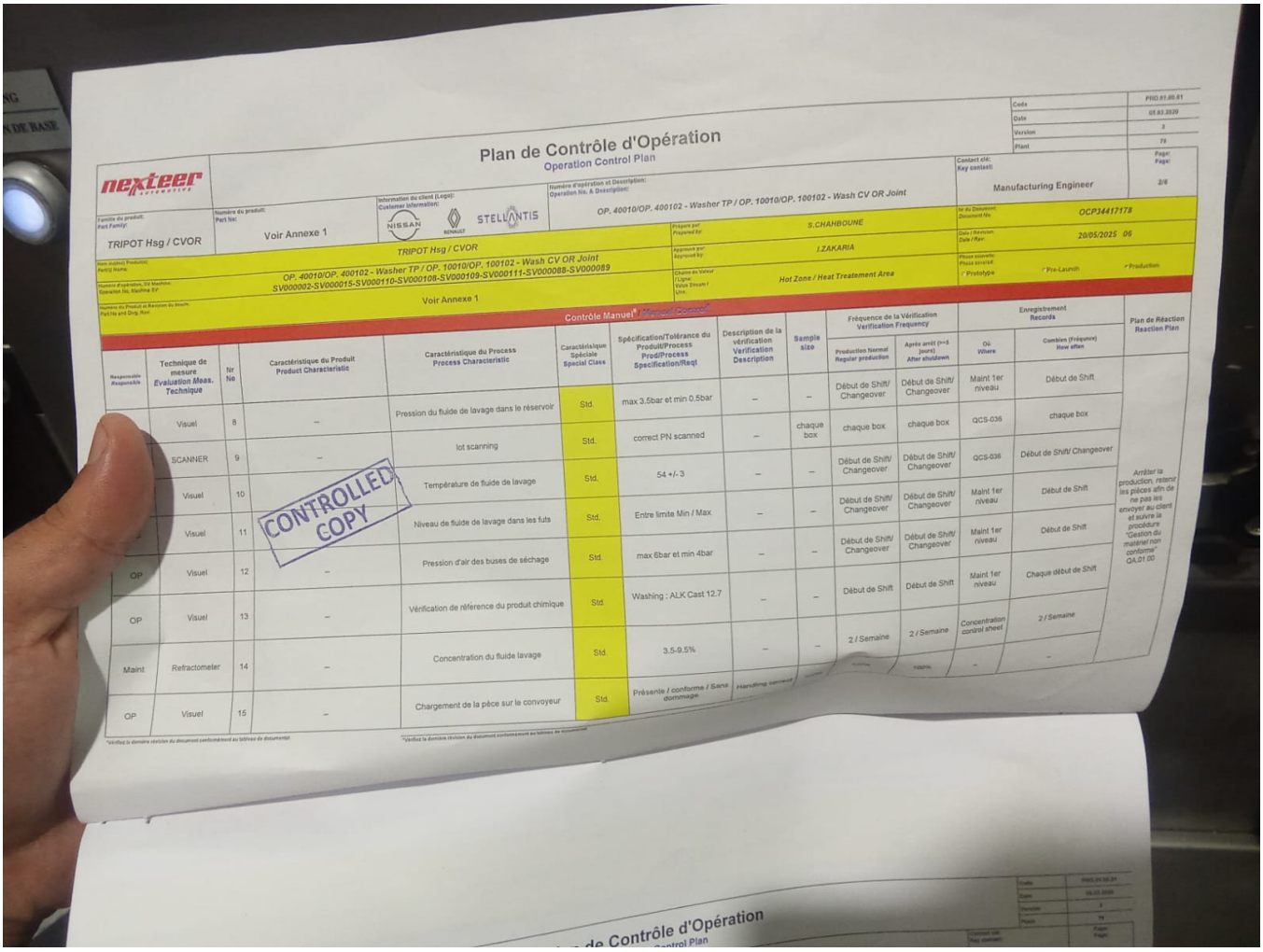
[24/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (3).jpeg**

Content: Interrupted Cycle (BLC-style) — reset, homing, wash entry, lamp start (upside-down photo)



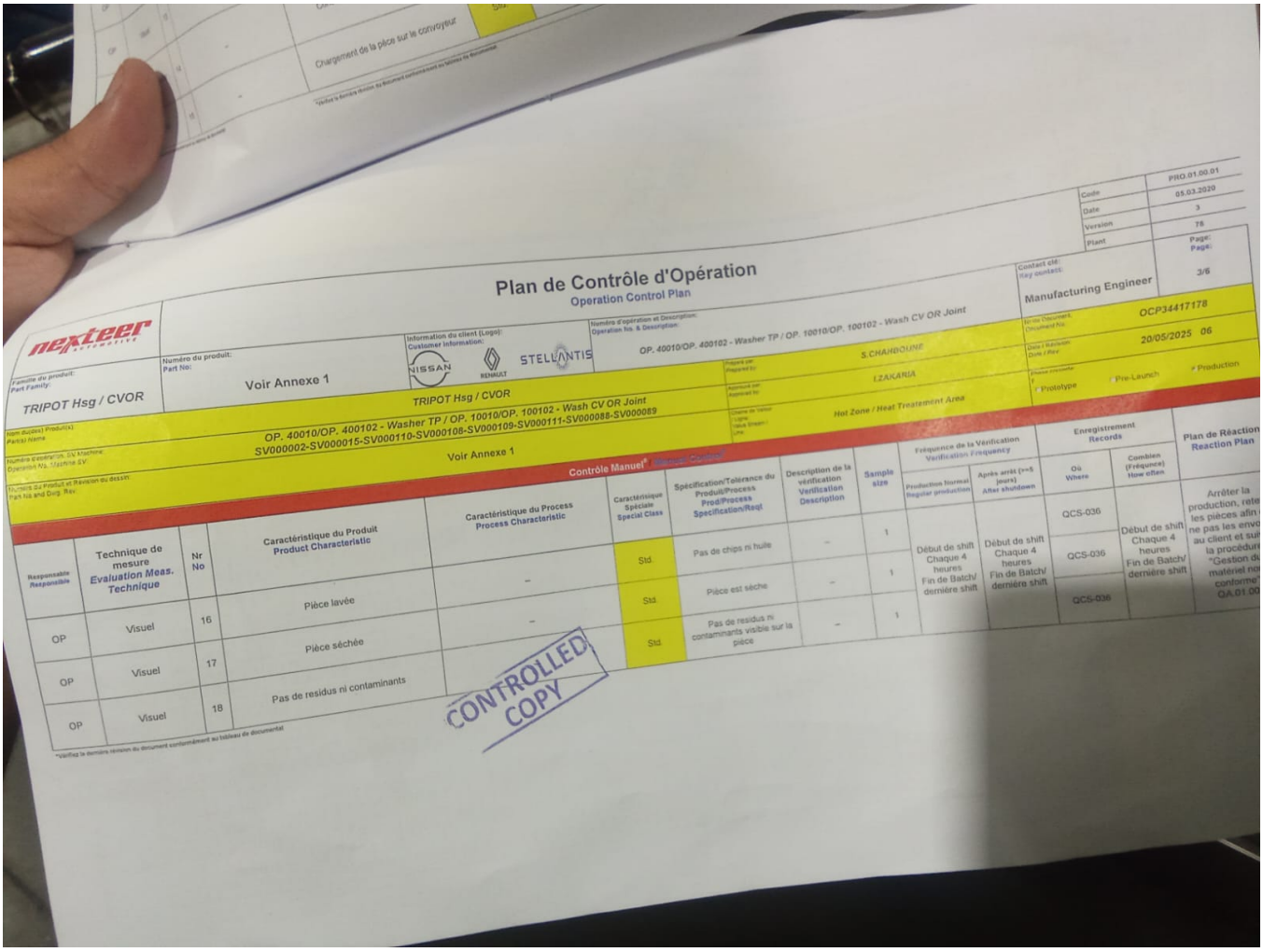
[25/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (4).jpeg**

Content: Operation Control Plan page 2 — manual checks 8–15



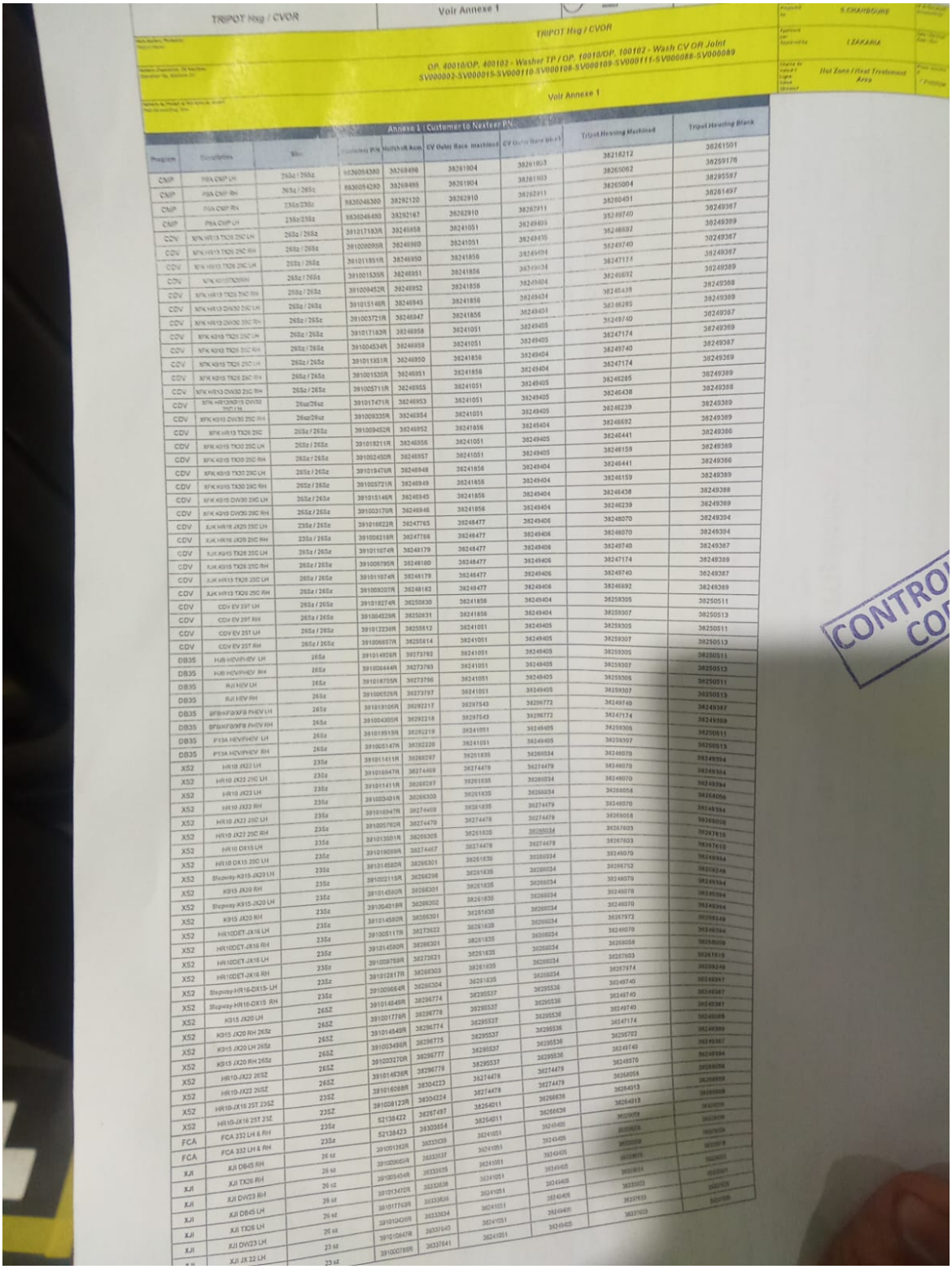
[26/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (5).jpeg**

Content: Operation Control Plan page 3 — checks 16–18 (washed, dried, no residue)



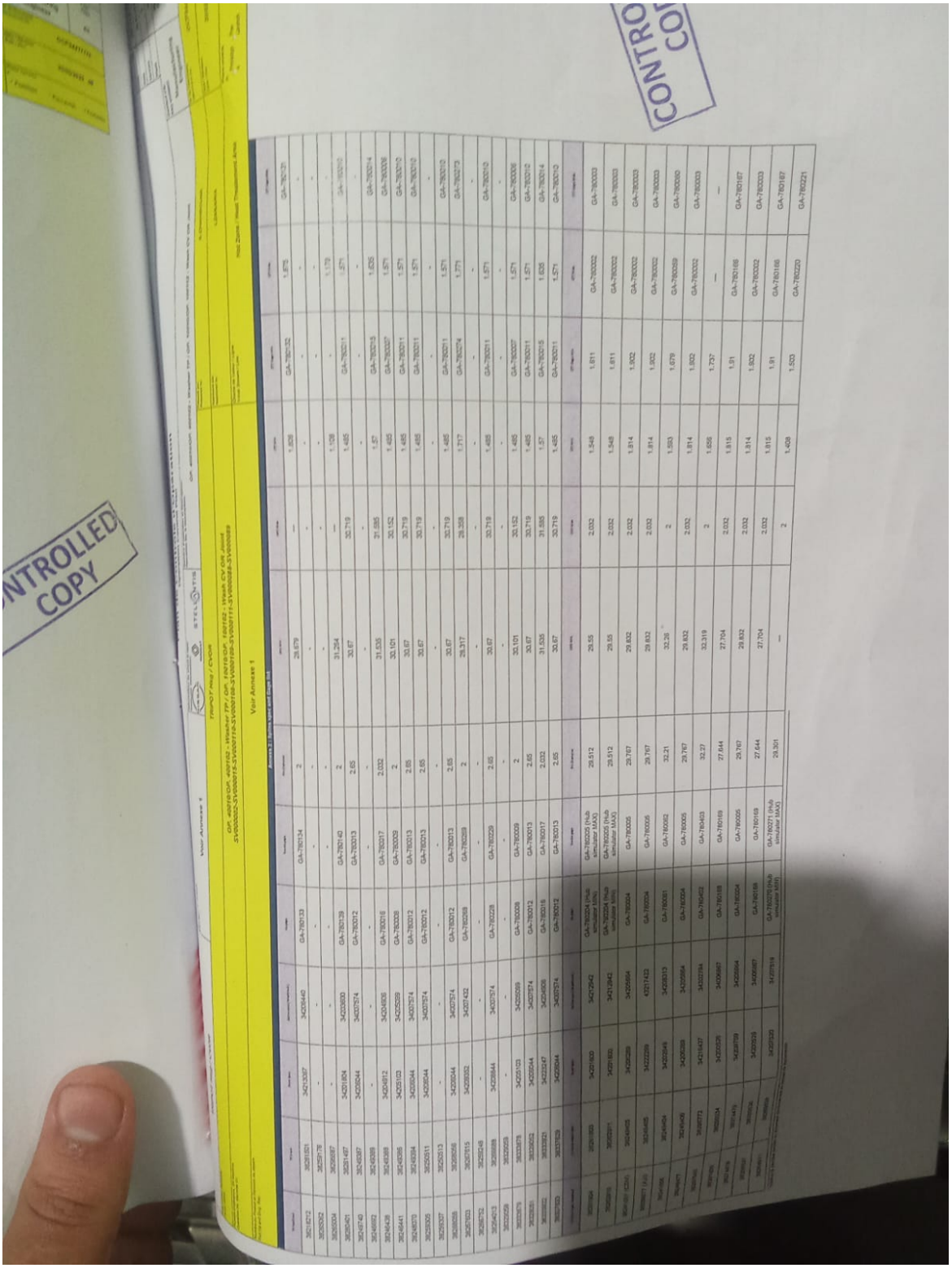
[27/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (6).jpeg**

Content: Annex 1 — Customer to Nexteer PN cross-reference (large table)



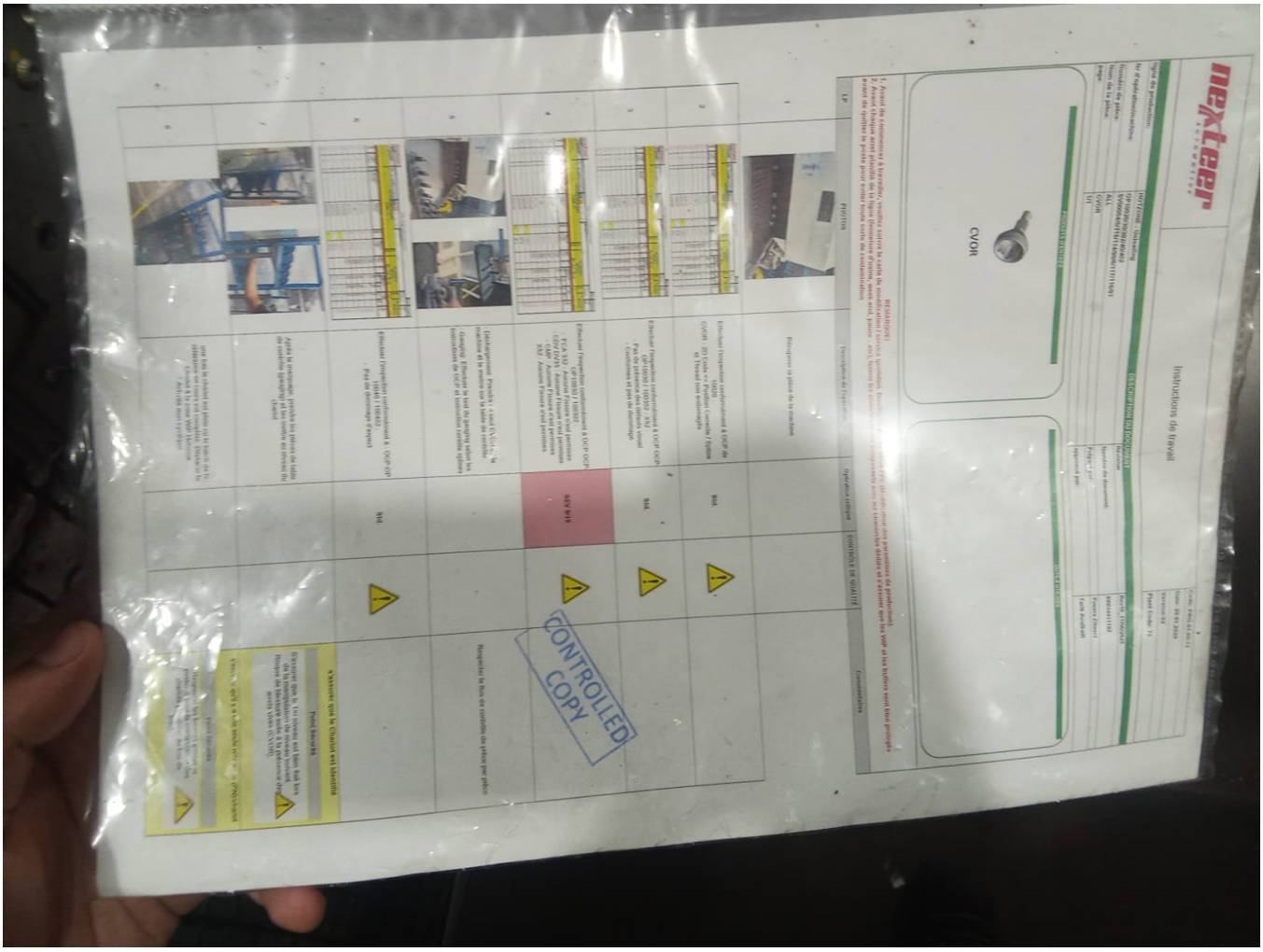
[28/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (7).jpeg**

Content: Annex 2 — Gage / fixture table (GA-780xxx pin numbers)



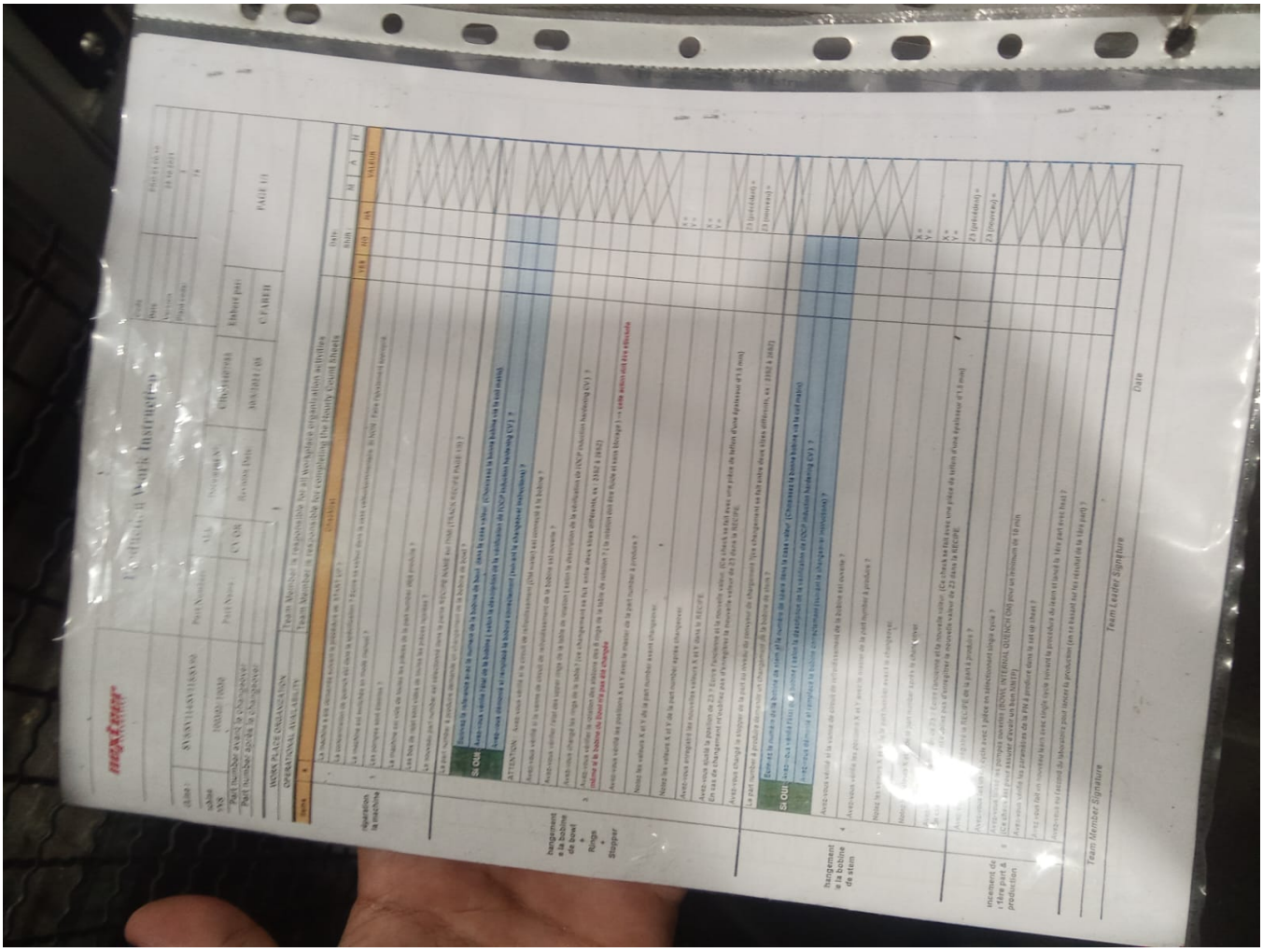
[29/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (8).jpeg**

Content: Instruction de Travail CVOR — full page with EP & OCP steps (landscape)



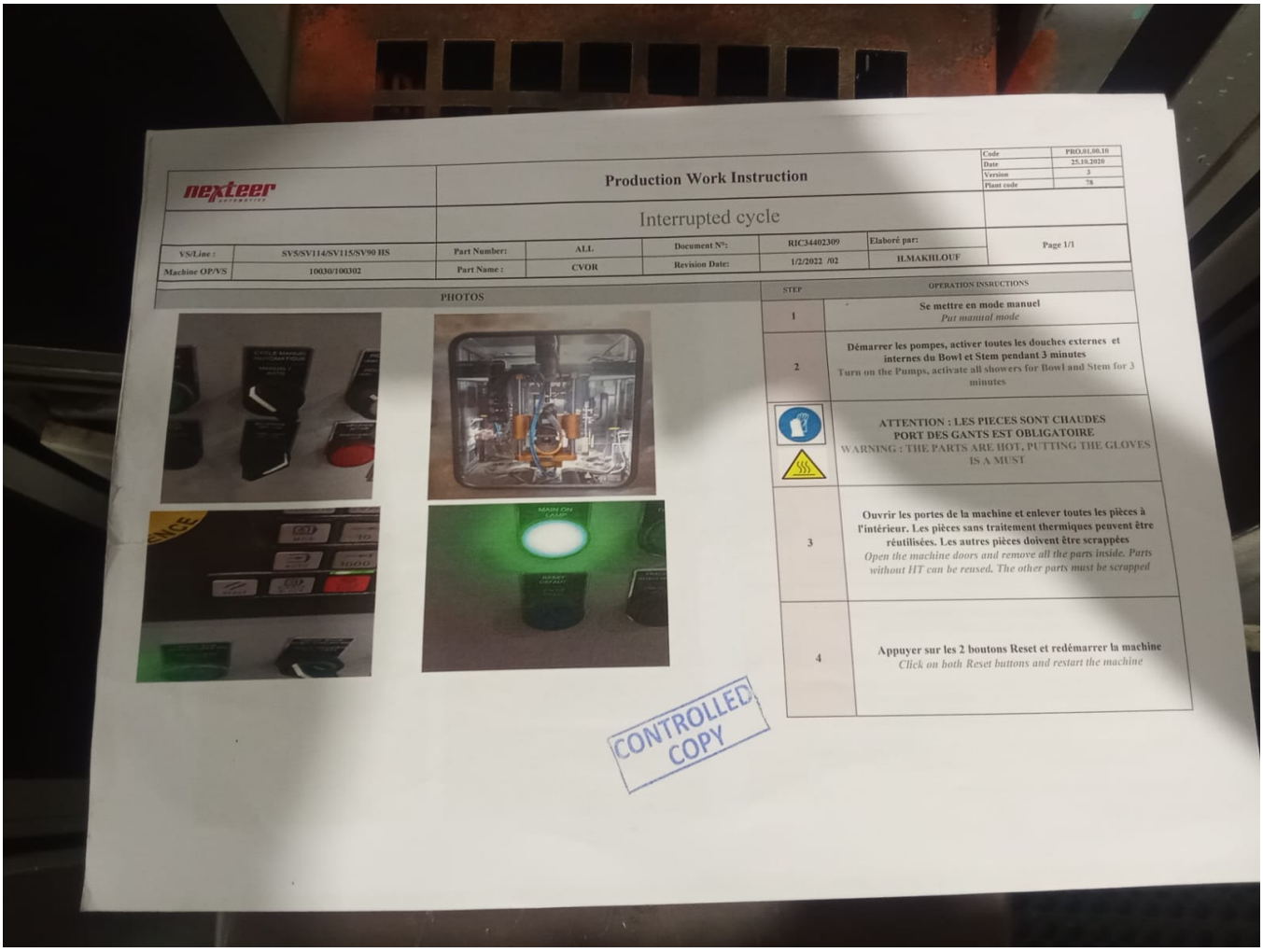
[30/64] **WhatsApp Image 2026-10-15 at 4.21.00 PM (9).jpeg**

Content: Production Work Instruction — Work Place Organization / 5S checklist (landscape)



[31/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM.jpeg**

Content: Supplementary Instructions — boxing orientation GOOD/BAD (red/green boxes, upside-down)



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Production Work Instruction

Interrupted cycle

Code	PRO.01.00.10
Date	25.10.2020
Version	3
Plant code	78

VS/Line :	SV5/SV114/SV115/SV90 HS	Part Number:	ALL	Document N°:	RIC34402309	Elaboré par:	Page 1/1
Machine OP/VS	10030/100302	Part Name :	CVOR	Revision Date:	1/2/2022 /02	IL.MAKHLOUF	

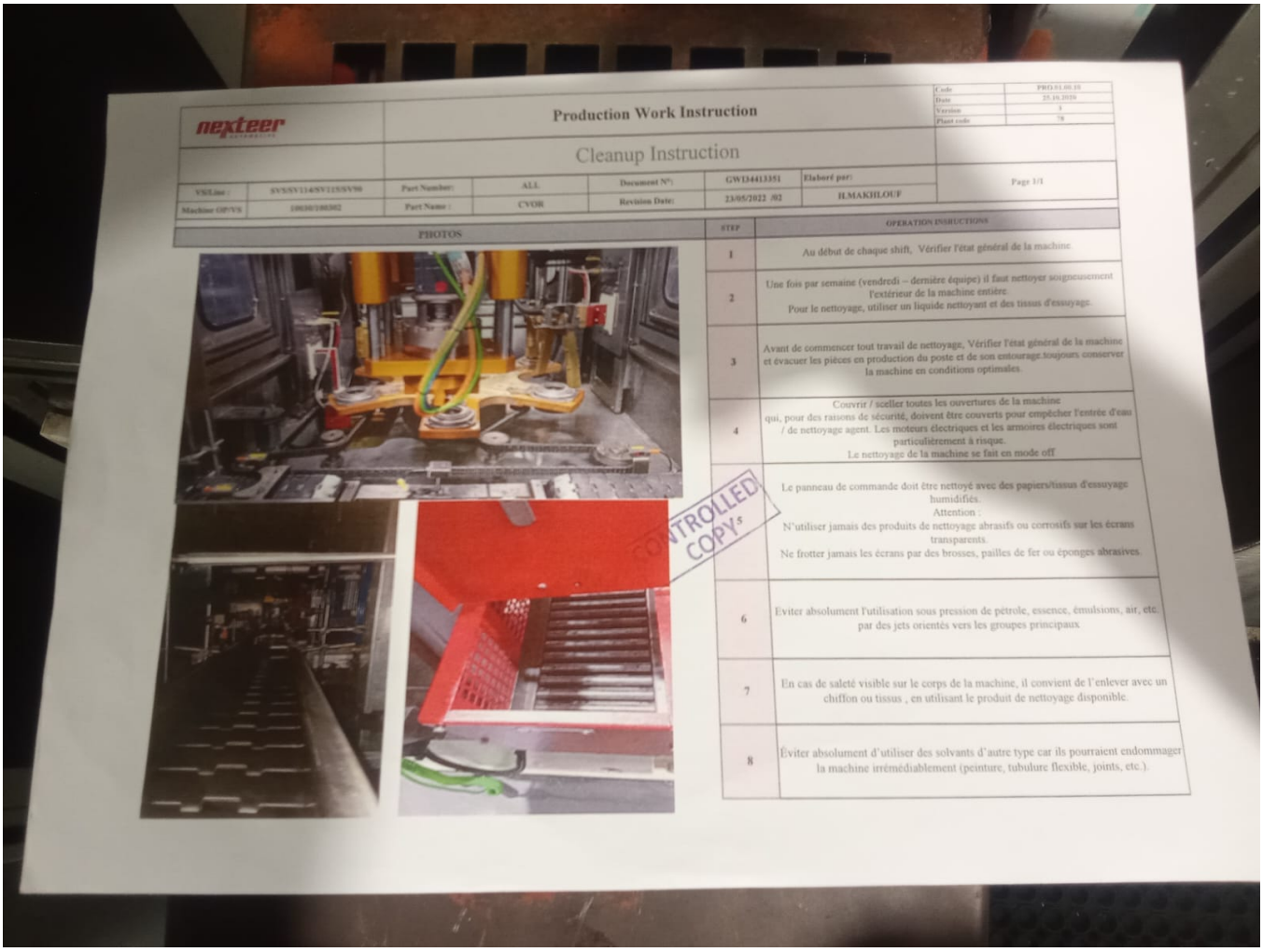
PHOTOS

STEP	OPERATION INSTRUCTIONS
1	**Se mettre en mode manuel** *Put manual mode*
2	**Démarrer les pompes, activer toutes les douches externes et internes du Bowl et Stem pendant 3 minutes** Turn on the Pumps, activate all showers for Bowl and Stem for 3 minutes
	ATTENTION : LES PIECES SONT CHAUDES PORT DES GANTS EST OBLIGATOIRE WARNING : THE PARTS ARE HOT, PUTTING THE GLOVES IS A MUST
3	**Ouvrir les portes de la machine et enlever toutes les pièces à l'intérieur. Les pièces sans traitement thermiques peuvent être réutilisées. Les autres pièces doivent être scrappées** *Open the machine doors and remove all the parts inside. Parts without HT can be reused. The other parts must be scrapped*
4	**Appuyer sur les 2 boutons Reset et redémarrer la machine** *Click on both Reset buttons and restart the machine*

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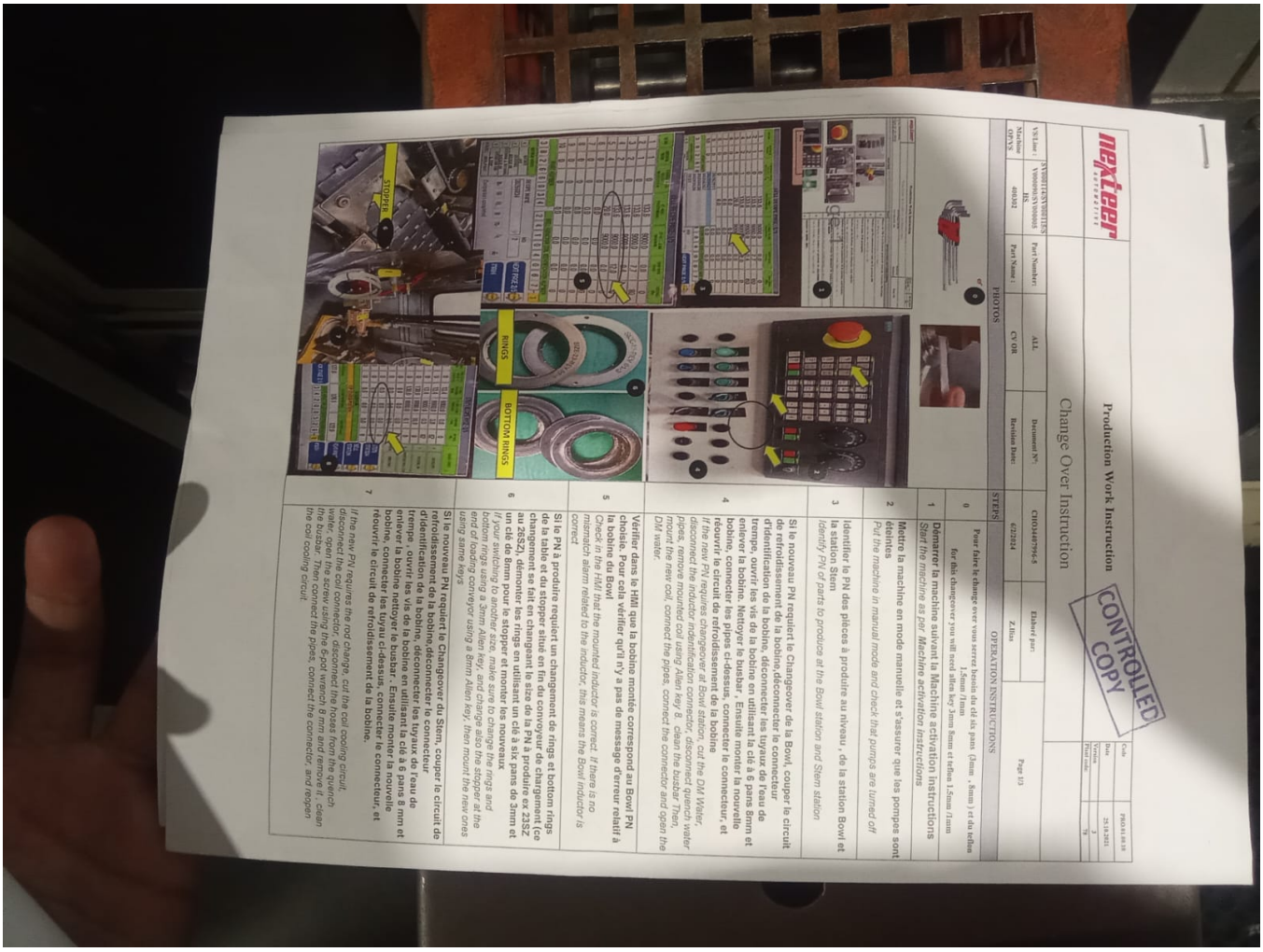
[32/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (1).jpeg**

Content: Cleanup Instruction (GWI34413351) — photos of induction machine interior



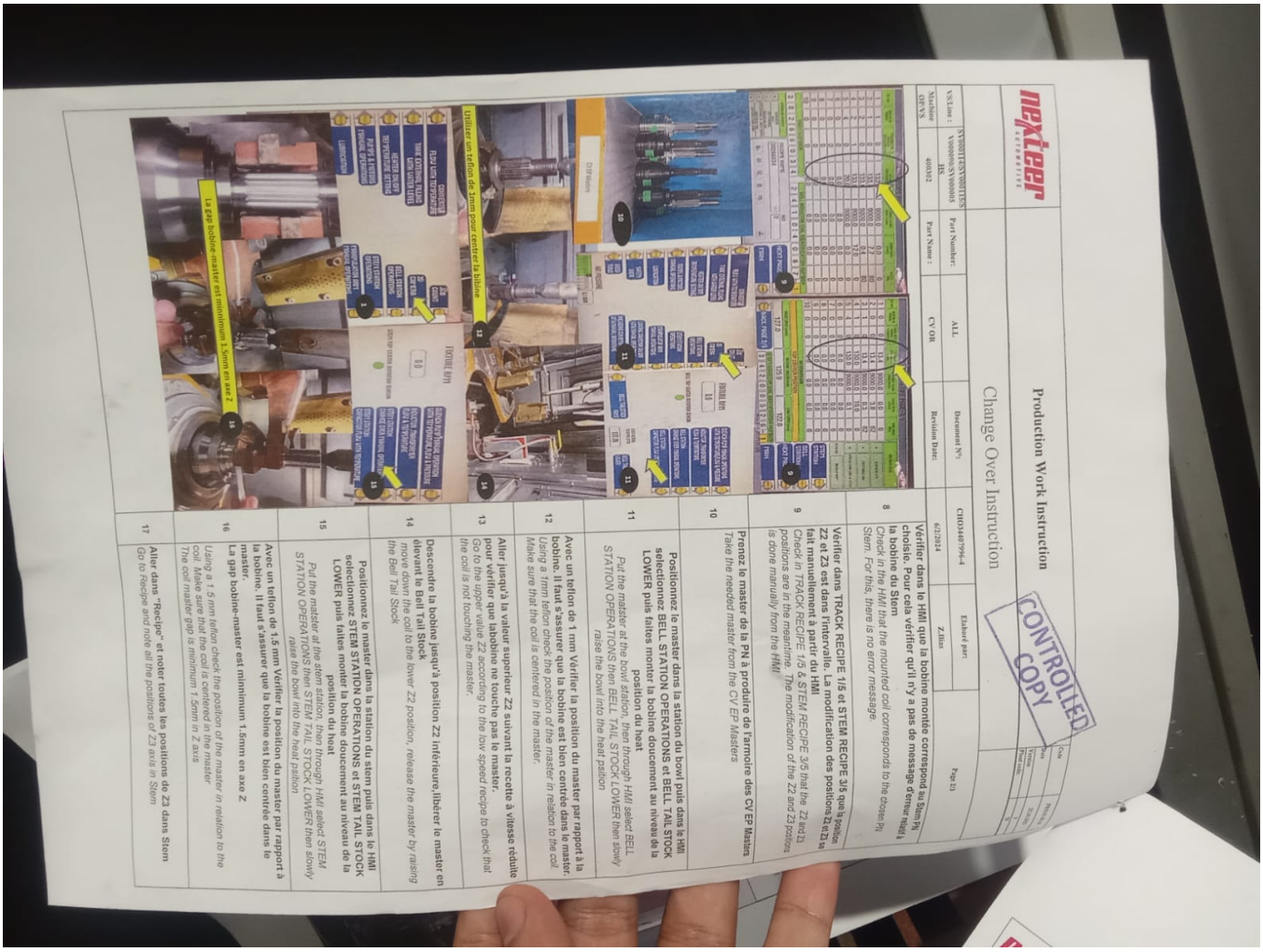
[33/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (2).jpeg**

Content: Sensibilité Check Instruction TP & CV (GWI3441351) — clip connections, inductor touch alarm



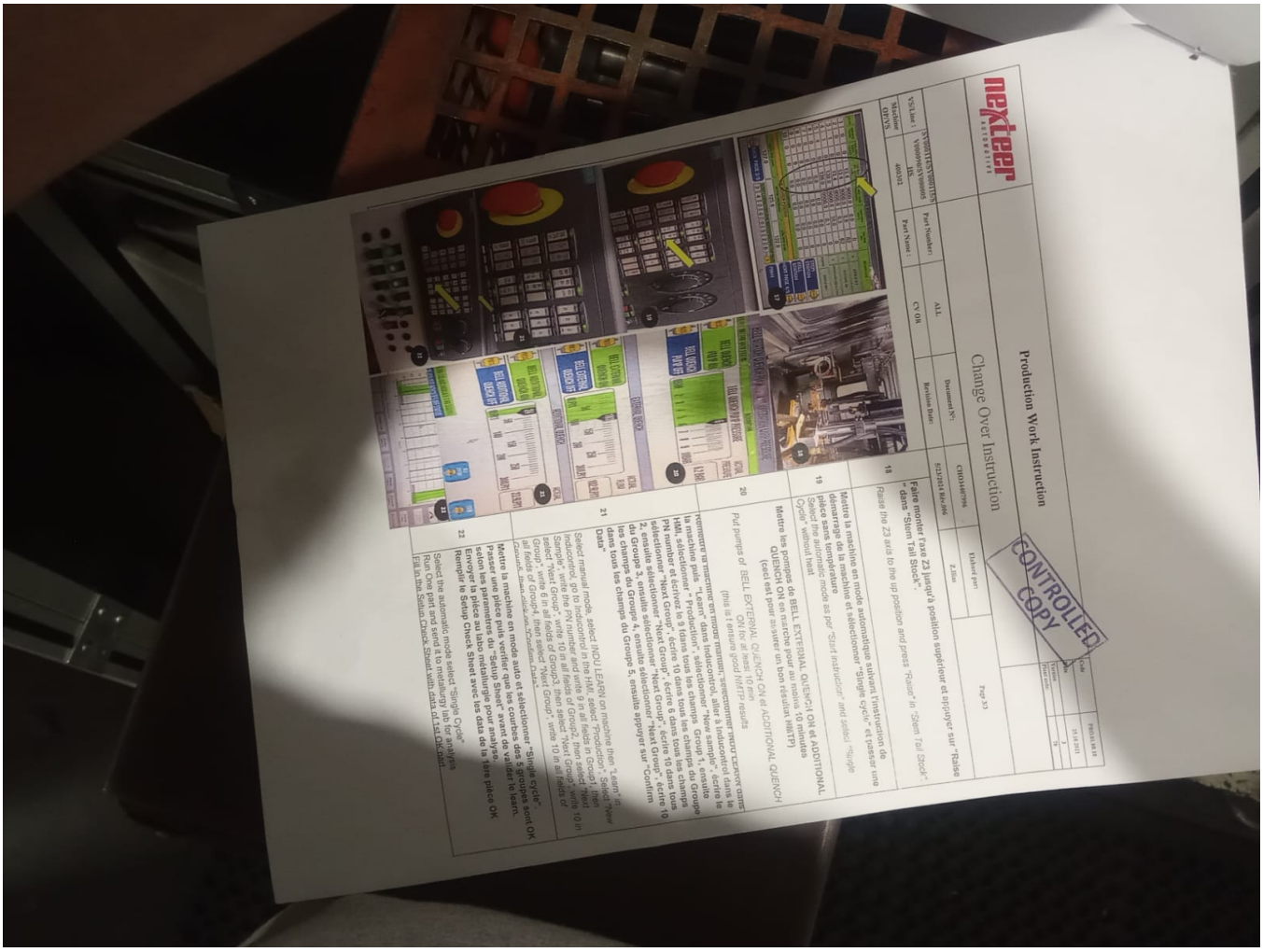
[34/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (3).jpeg**

Content: Change Over Instruction (CHO3440796-4) page 13 — coil, rings, bottom rings, Allen keys



[35/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (4).jpeg**

Content: Change Over page 23 — master placement, teflon gap, Z2/Z3, Bell/STEM recipes



[36/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (5).jpeg**

Content: Change Over page 23 (cont.) — Learn on Inducontrol, groups, single cycle, setup sheet



nexteer AUTOMOTIVE

EP VERIFICATION INSTRUCTIONS-EP

Code: PRO.01.00.11
Date:22.09.2021
Version: 03
Plant code: 78

VS/LINE	SV5/SV90/SV114/SV115	PART NUMBER	CV OR	DOCUMENT NO.	EPR34409900	PAGE
MACHINE OP	OP_100302/10030	PART NAME	All	REVISION DATE/NO.	01-07-2025_006	1/2

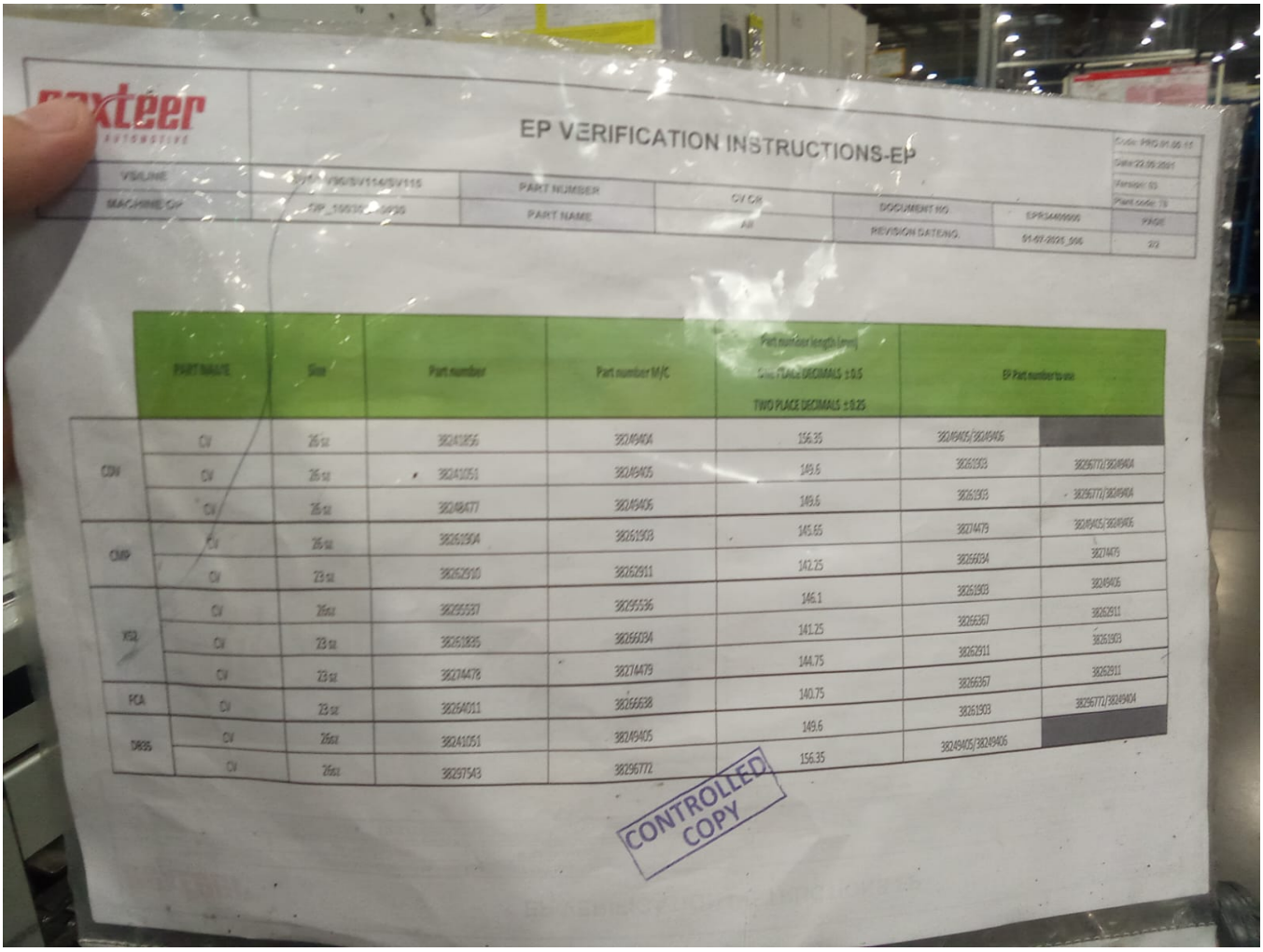
PHOTOS

CONTROLLED COPY

N°	OPERATION INSTRUCTION
	Error proofing for geometry of the part
1	***Démarrer la machine en suivant l'instruction de démarrage de la machine et mettre la machine en mode automatique.*** *Start the machine as per "Start instruction" and select automatic mode.*
2	***Lancer le test avec les pièces master définies au tableau en page 2/2 et s'assurer que les pièces sont rejetées.*** *Run the test with the master parts as defined in the table in page 2/2 and check that the parts are rejected .*
	Error proofing for 2D code
1	***Lancer une pièce (part blank) avec le même PN que celui séléctioné au Recipe et s'assurer que la pièce n'est pas rejetée.Scanner le 2D code de la pièce et s'assurer que c'est le PN requis*** *Run the test with the master part that selected in the Recipe and check that the parts is not rejected . Scan the 2D code and check that the PN is correct.*
2	***Lancer une pièce sans 2D code (part blank) en désactivant le marquage (laser desable) et s'assurer que la pièce est rejetée.*** *Run a part without 2D code (blank part) with engraver desabled (laser desable) and check that the part is rejected.*
3	***Lancer la pièce master avec le 2D code endommagé en désactivant le marquage (laser desable) et s'assurer que la pièce est rejetée.*** *Run the master part with damaged 2D code with engraver desabled (laser desable) and check that the part is rejected.*
4	***Lancer une pièce déjà traitée du même PN séléctioné au Recipe et s'assurer que la pièce est rejetée.*** *Run a heat-treated part with the same PN selected in the Recipe and check that the part is rejected.*
	La vérification EP se fait à l'entrée de la station induction avant le (DataMatrix). EP verification to be done in the entrance of induction station before (DataMatrix).
	Les pièces nécessaires pour la vérification EP doivent être mises à disposition au poste de travail au préalable par le team leader. Required parts for EP verification shall be made available in work station in advance by the team leader.
	EP FAILURE – CONDUCT
	Informer le Team Leader, faire appel au ME pour corriger l'erreur. Ne commencer la production que la vérification de l'EP st positive. *Inform the Team Leader, call ME to remove the failure. Production can be started following positive result of EP verification.*

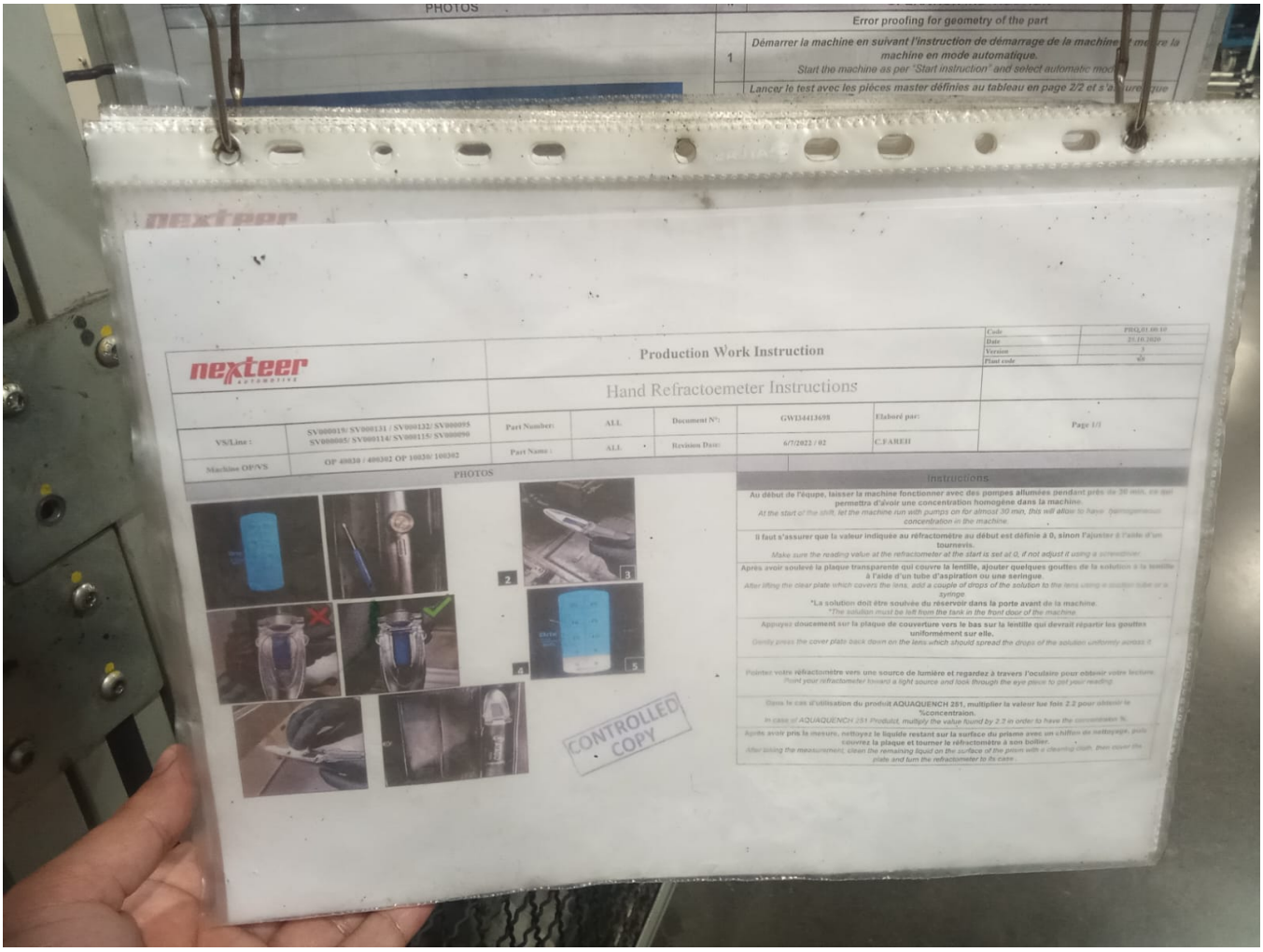
[37/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (6).jpeg**

Content: EP Verification Instructions (EPR34409900) page 1/2 — geometry & 2D code error proofing



[38/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (7).jpeg**

Content: EP Verification page 2/2 — master part numbers by program (CDV, CMP, XS2, FCA, DB35)



[39/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (8).jpeg**

Content: Hand Refractometer Instructions (GWI34413698) — sampling, ×2 Aquaquench 251



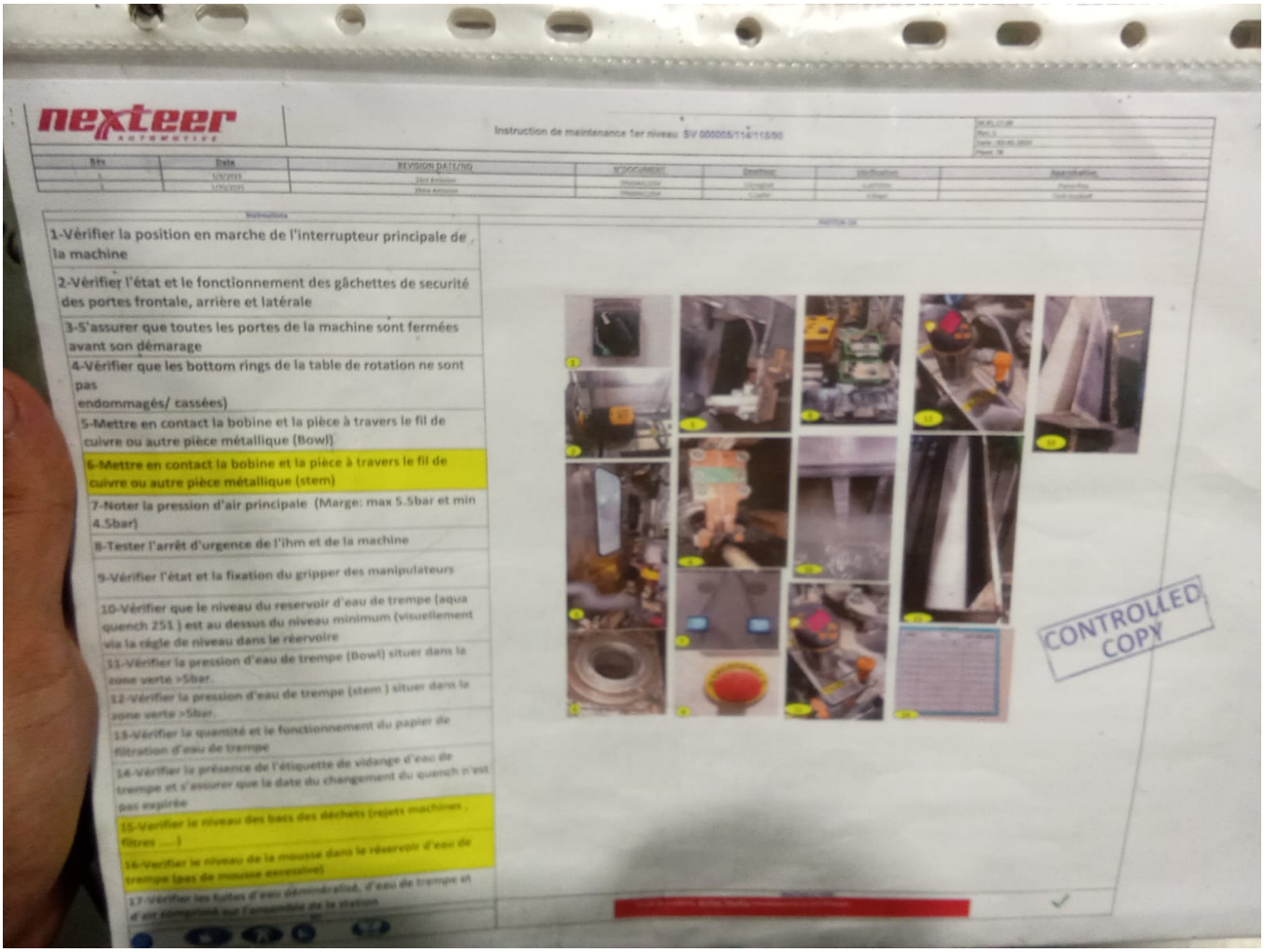
[40/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (9).jpeg**

Content: Supplementary / Crack Reaction Plan (GWI34417543) — 14 checks + systemic actions

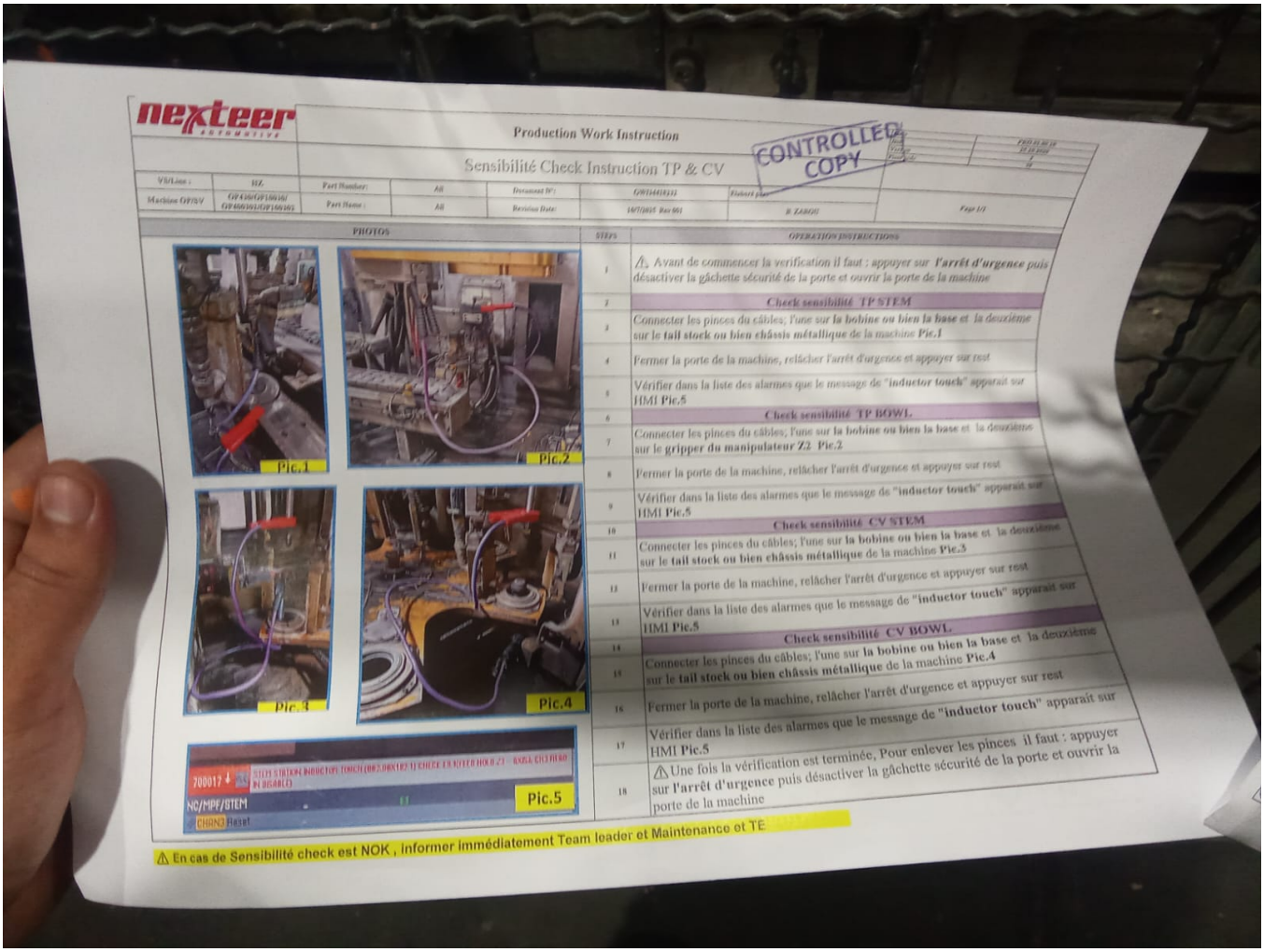


[41/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (10).jpeg**

Content: 1st Level Maintenance (alt. SV 000006/114/115/90) — 17 steps, quench water, foam check

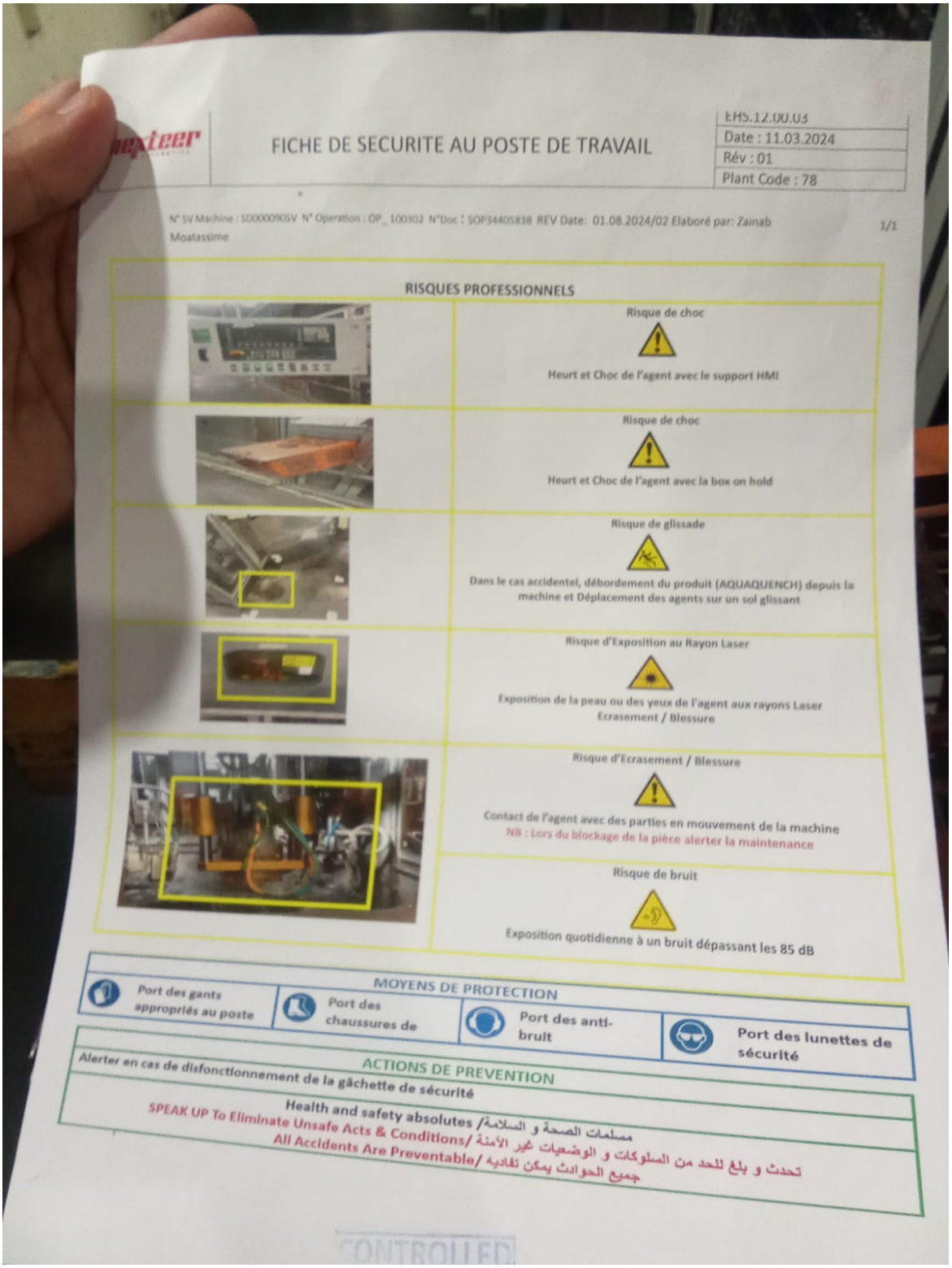


[42/64] **WhatsApp Image 2026-10-15 at 4.21.01 PM (11).jpeg**



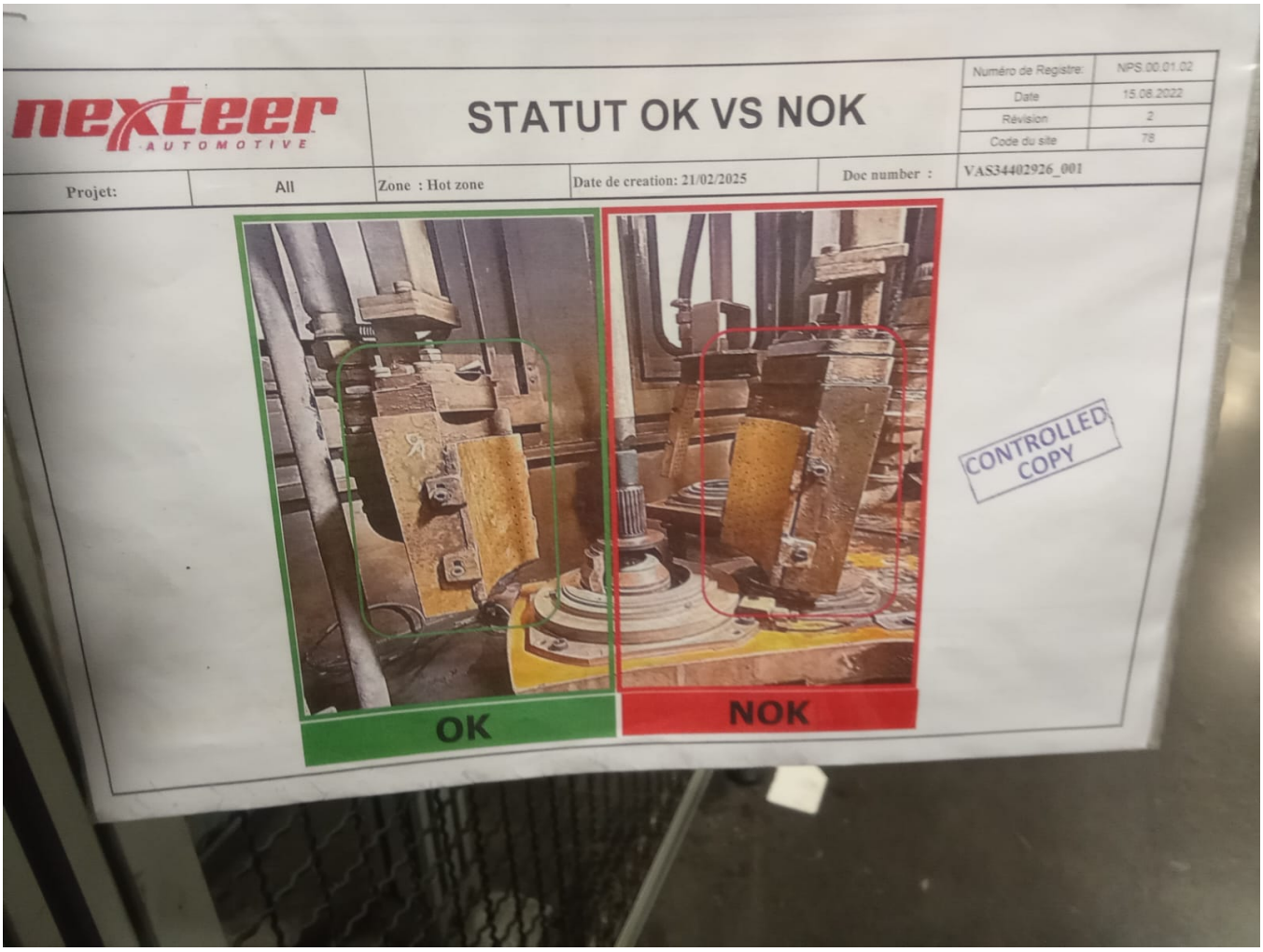
[43/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM.jpeg**

Content: Inducontrol HMI — shower flow graph, Rejected status, threshold 300 L/min



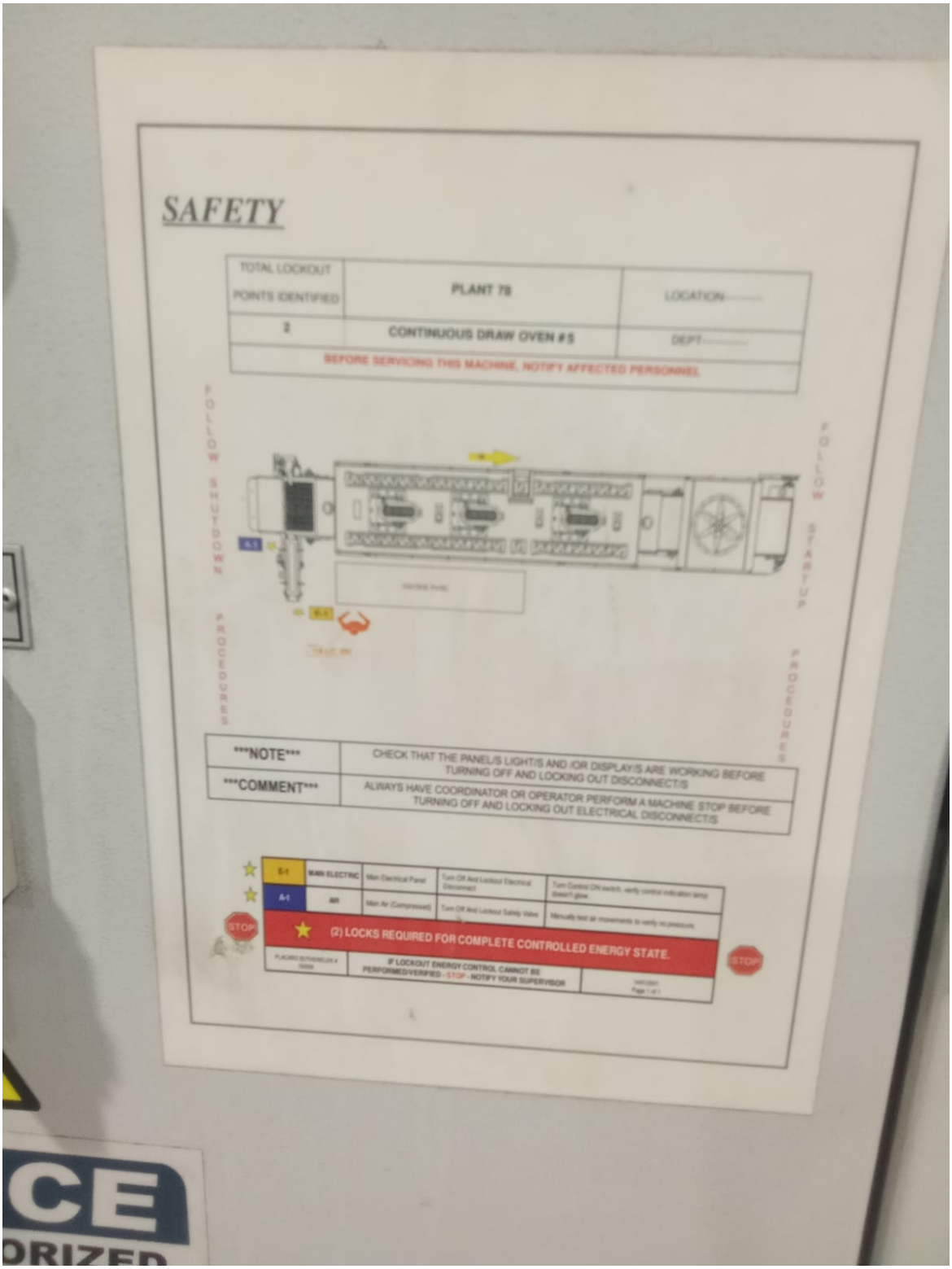
[44/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (1).jpeg**

Content: Interrupted Cycle (RIC34402309) — manual mode, 3-min showers, hot-part glove warning



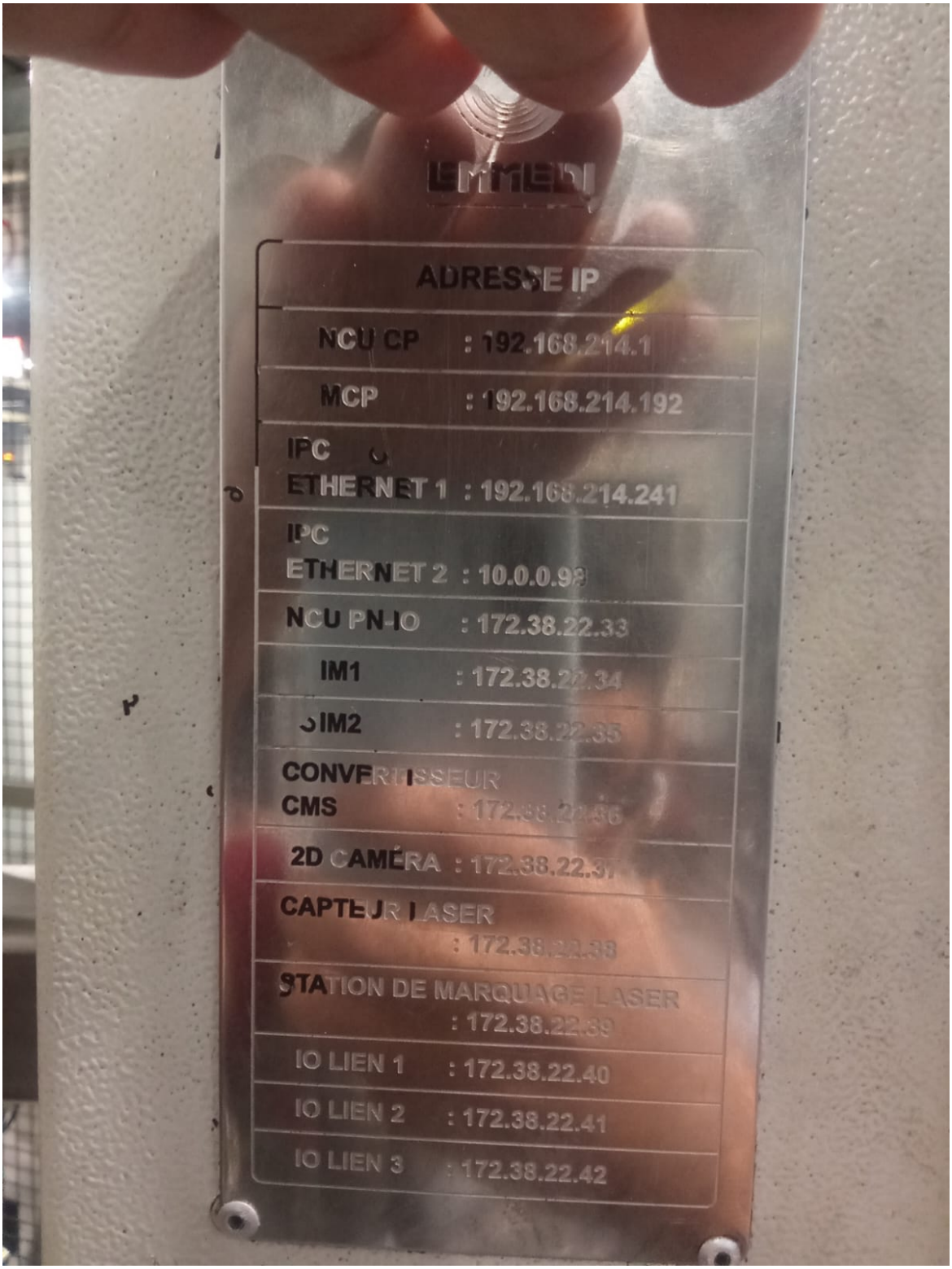
[45/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (2).jpeg**

Content: OK vs NOK — inductor coil alignment



[46/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (3).jpeg**

Content: Interrupted Cycle (RIC34401371) — clear errors, Safety Reset, Auto-Start ×2



[47/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (4).jpeg**

Content: Safety LOTO placard — Draw Oven #5, E-1 + A-1, 2 locks required



[48/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (5).jpeg**

Content: OK vs NOK — part seating on fixture (straight vs tilted)



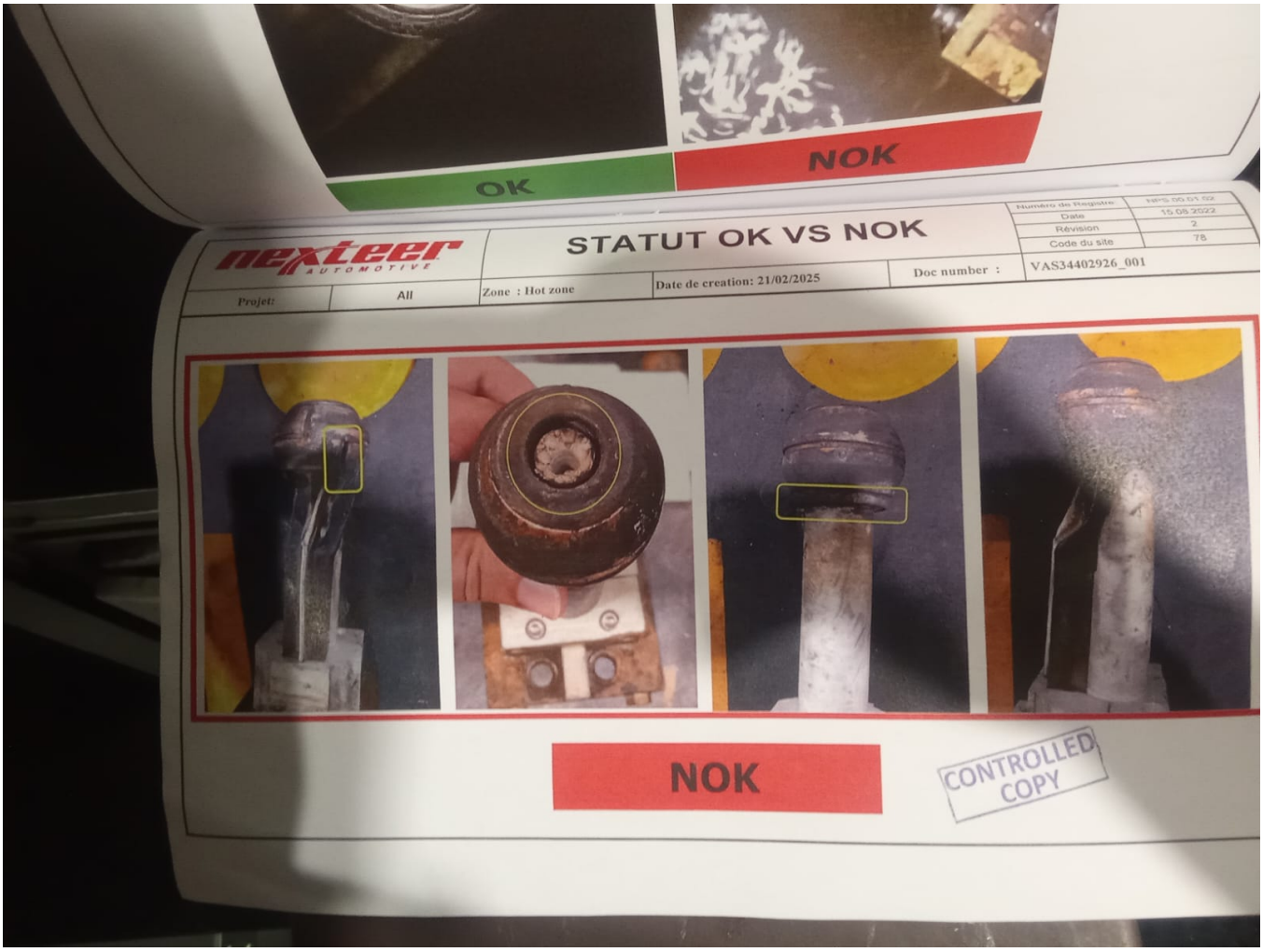
[49/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (6).jpeg**

Content: OK vs NOK — bell quench surface condition



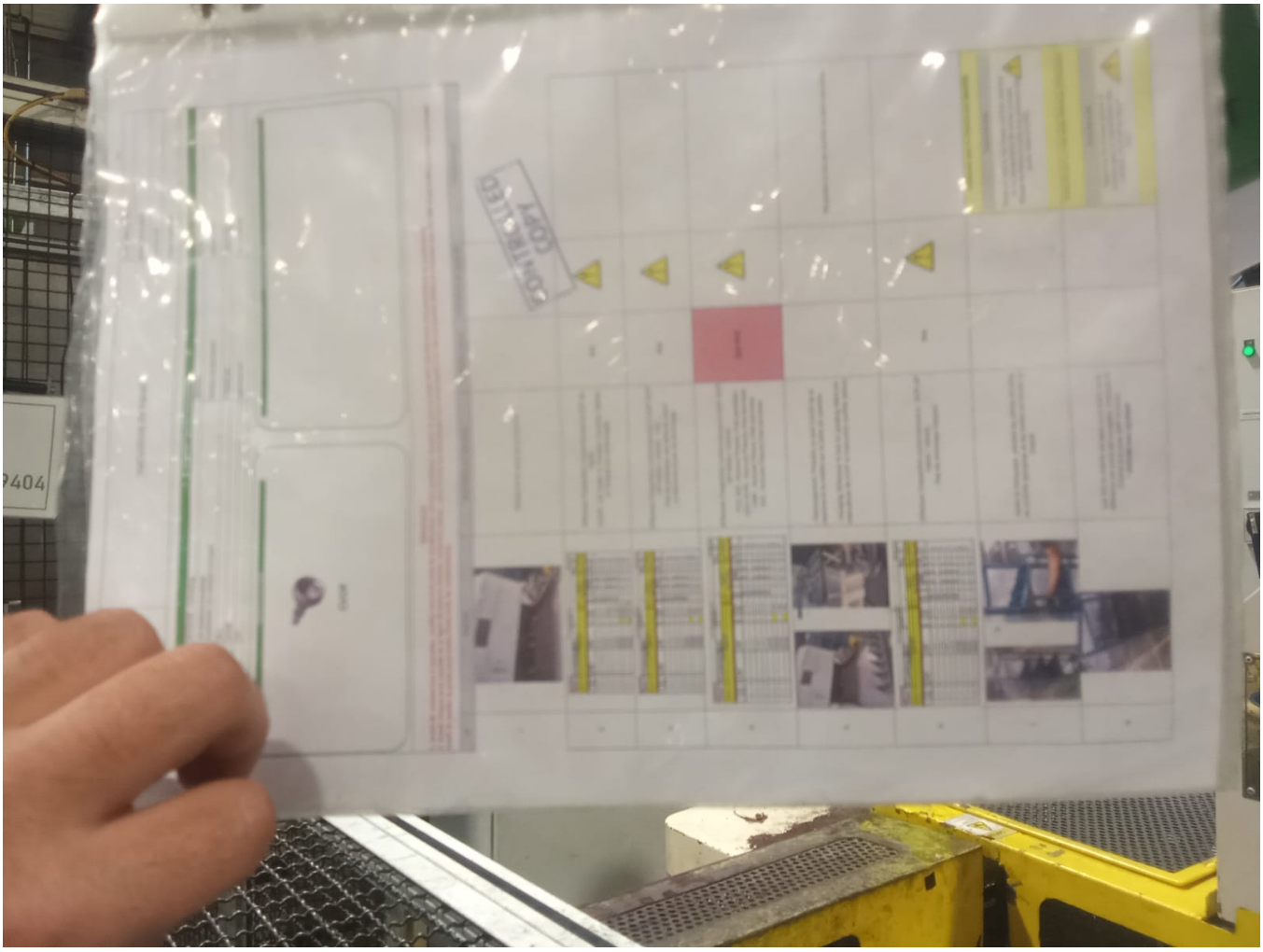
[50/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (7).jpeg**

Content: OK vs NOK — heated part / scale defects



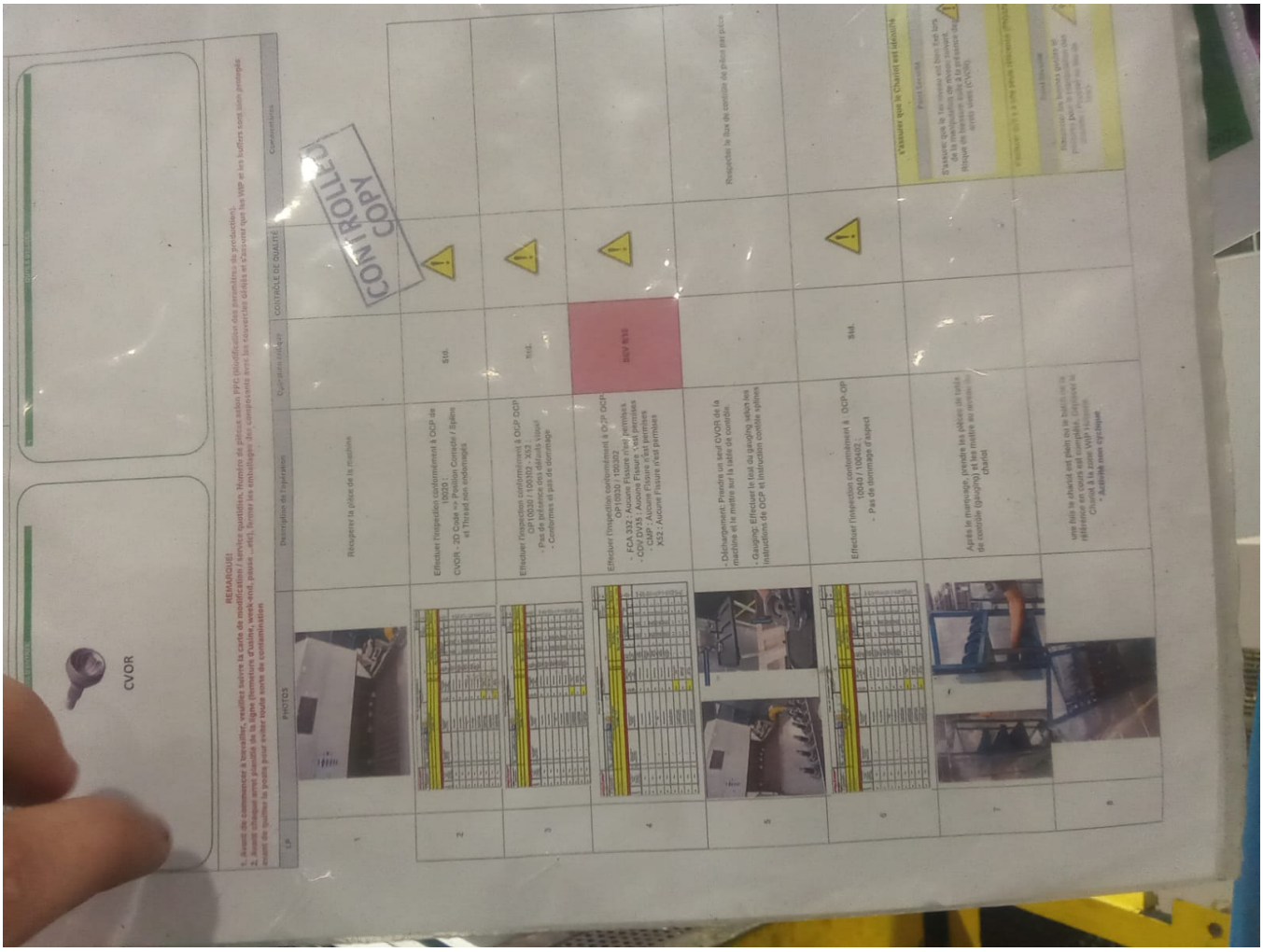
[51/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (8).jpeg**

Content: Fiche de Sécurité au Poste de Travail (SOP34405838) — risks & PPE



[52/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (9).jpeg**

Content: Fiche de Sécurité (alt. orientation) — same risk matrix



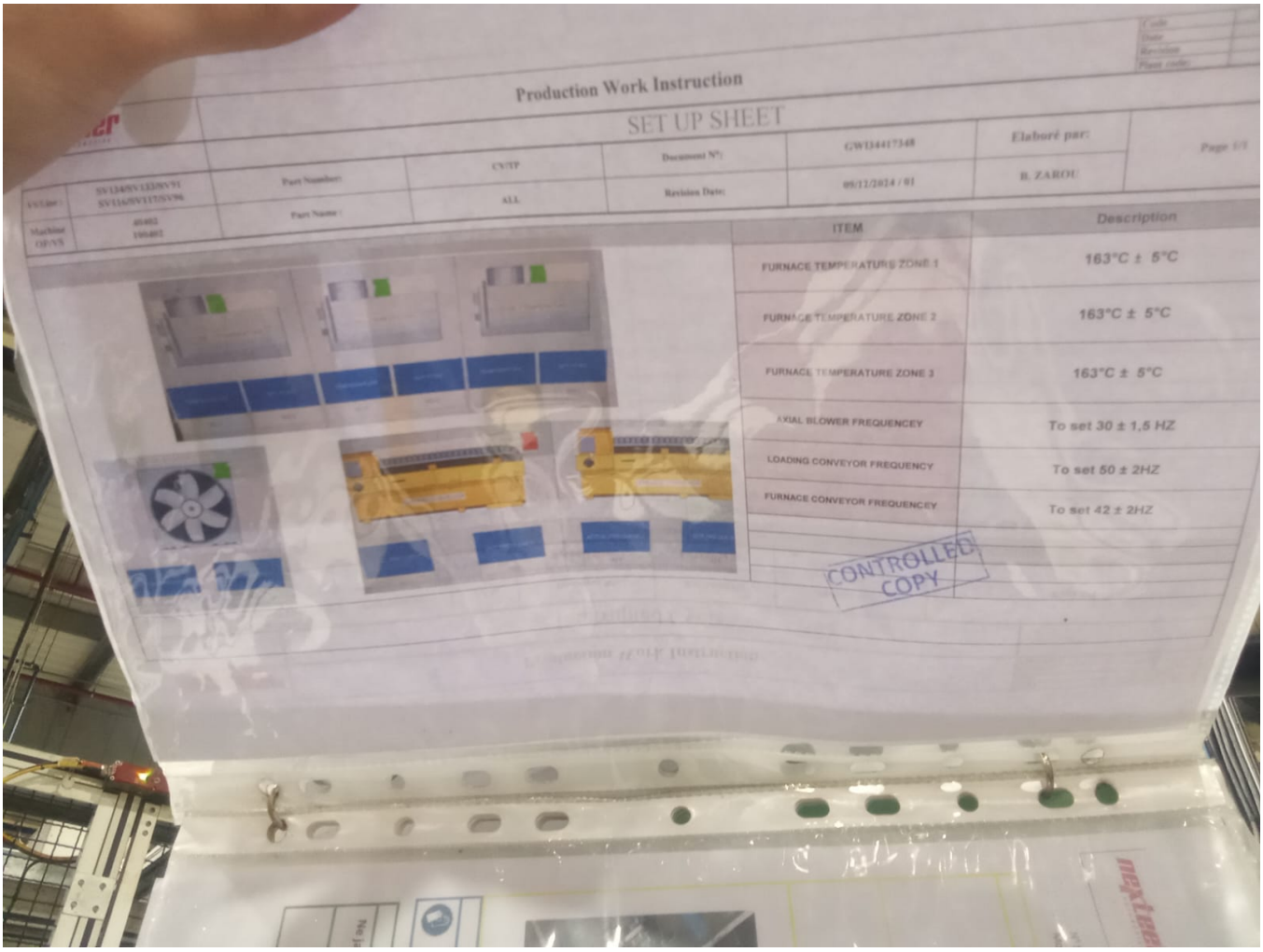
[53/64] **WhatsApp Image 2026-10-15 at 4.21.02 PM (10).jpeg**

Content: IP address plate (EMMEPI) — full network table



[54/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM.jpeg**

Content: Nexteer Automotive main HMI — Traceability / Operations / Recipe buttons, 3D line model



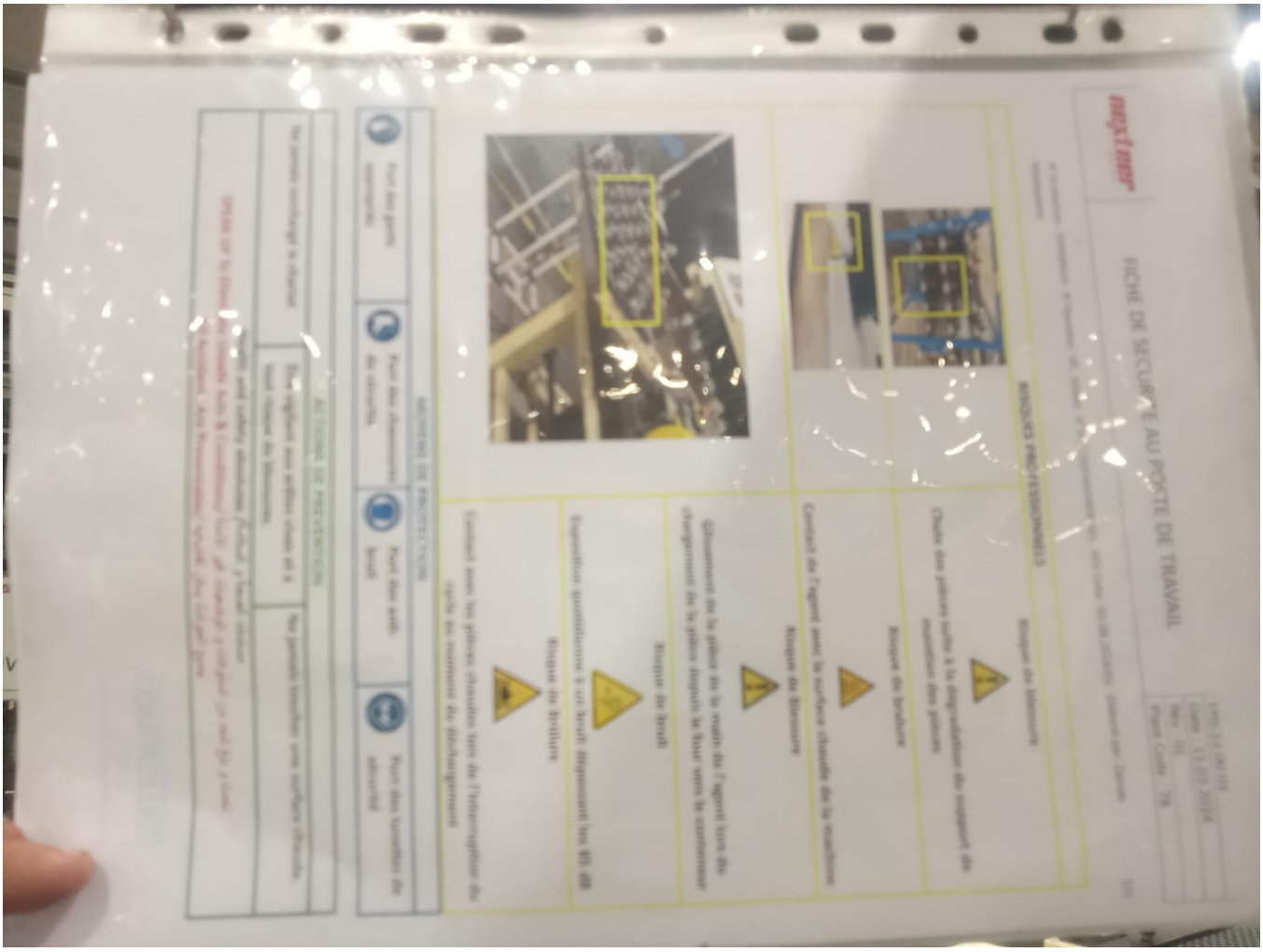
[55/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (1).jpeg**

Content: Non-conforming material handling flow chart (landscape, blurry)



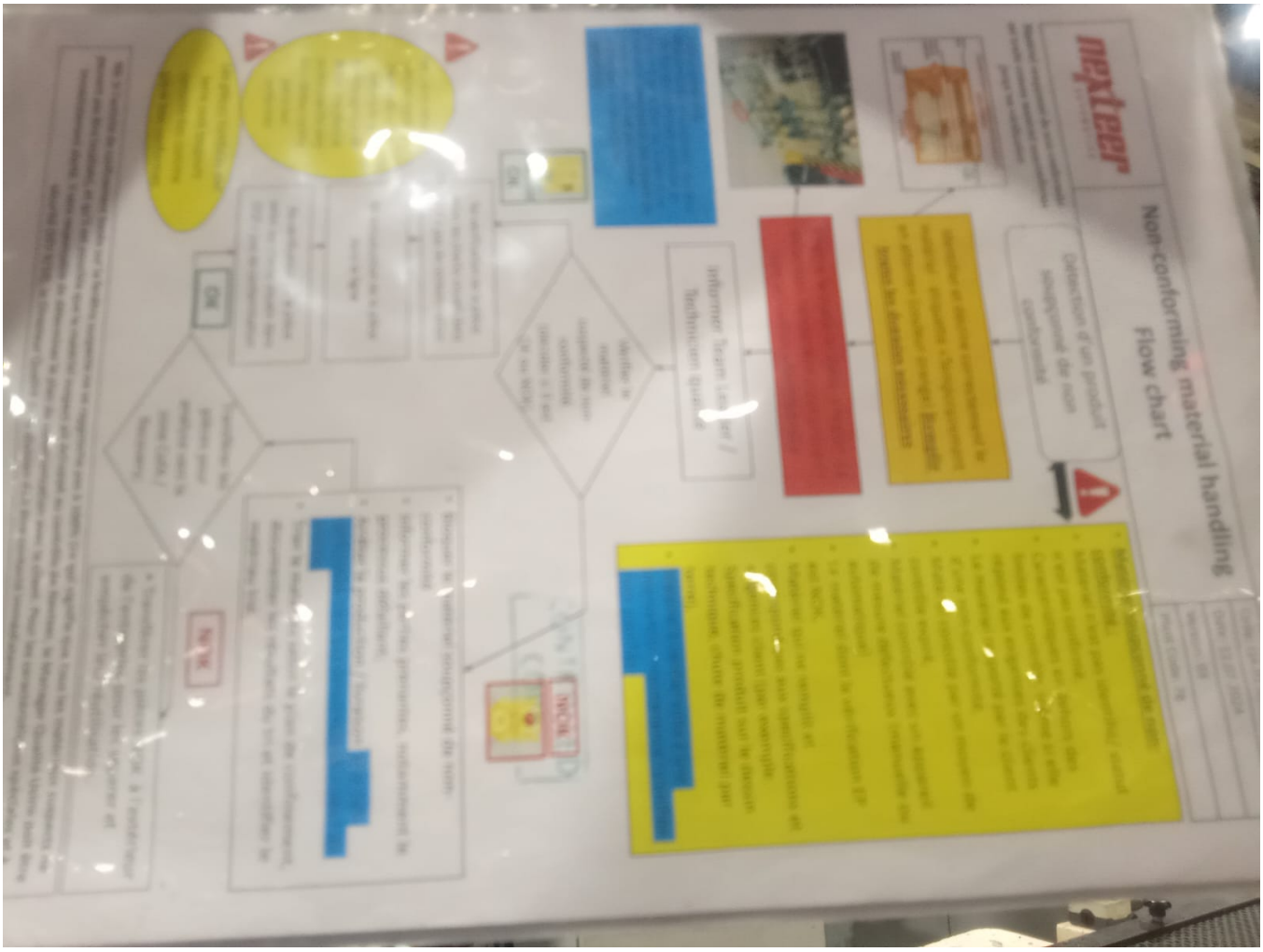
[56/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (2).jpeg**

Content: Gage Instructions + Non-conforming flow (two pages in sleeve)



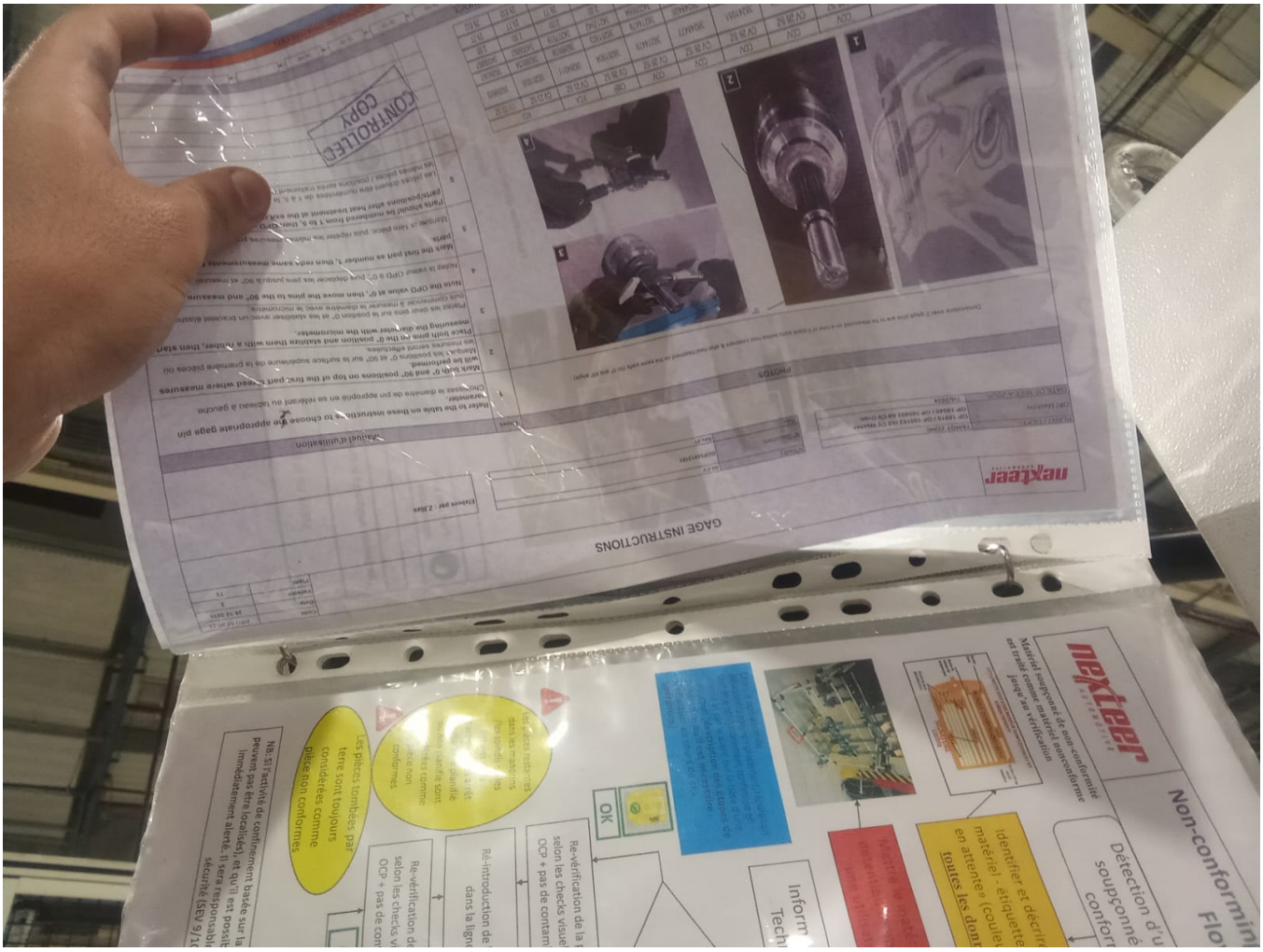
[57/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (3).jpeg**

Content: Non-conforming material handling flow chart (clearer)



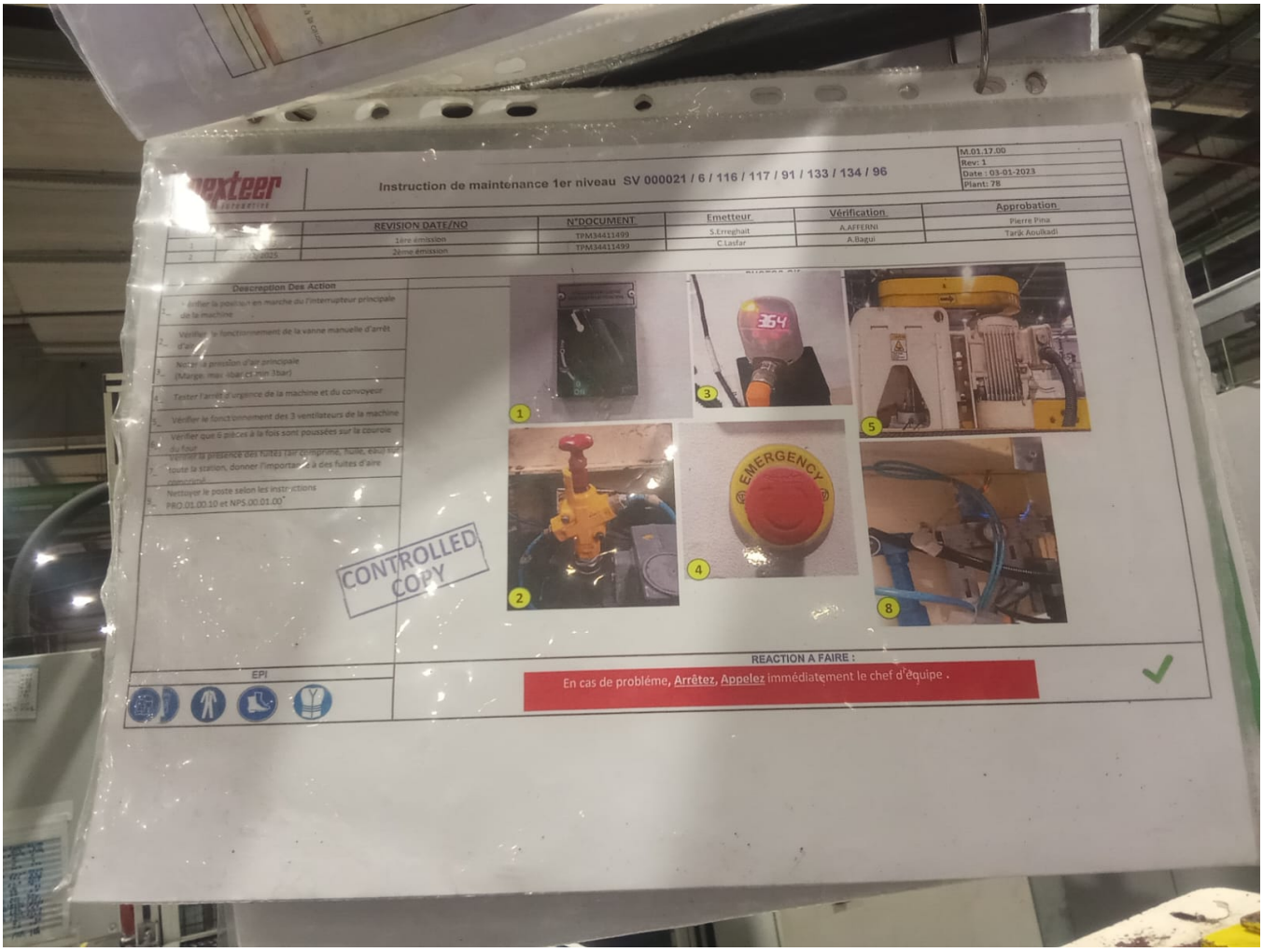
[58/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (4).jpeg**

Content: 1st Level Maintenance (TPM34411499) — 8 steps, air valve, E-stop, 3 fans, 6-pcs belt check



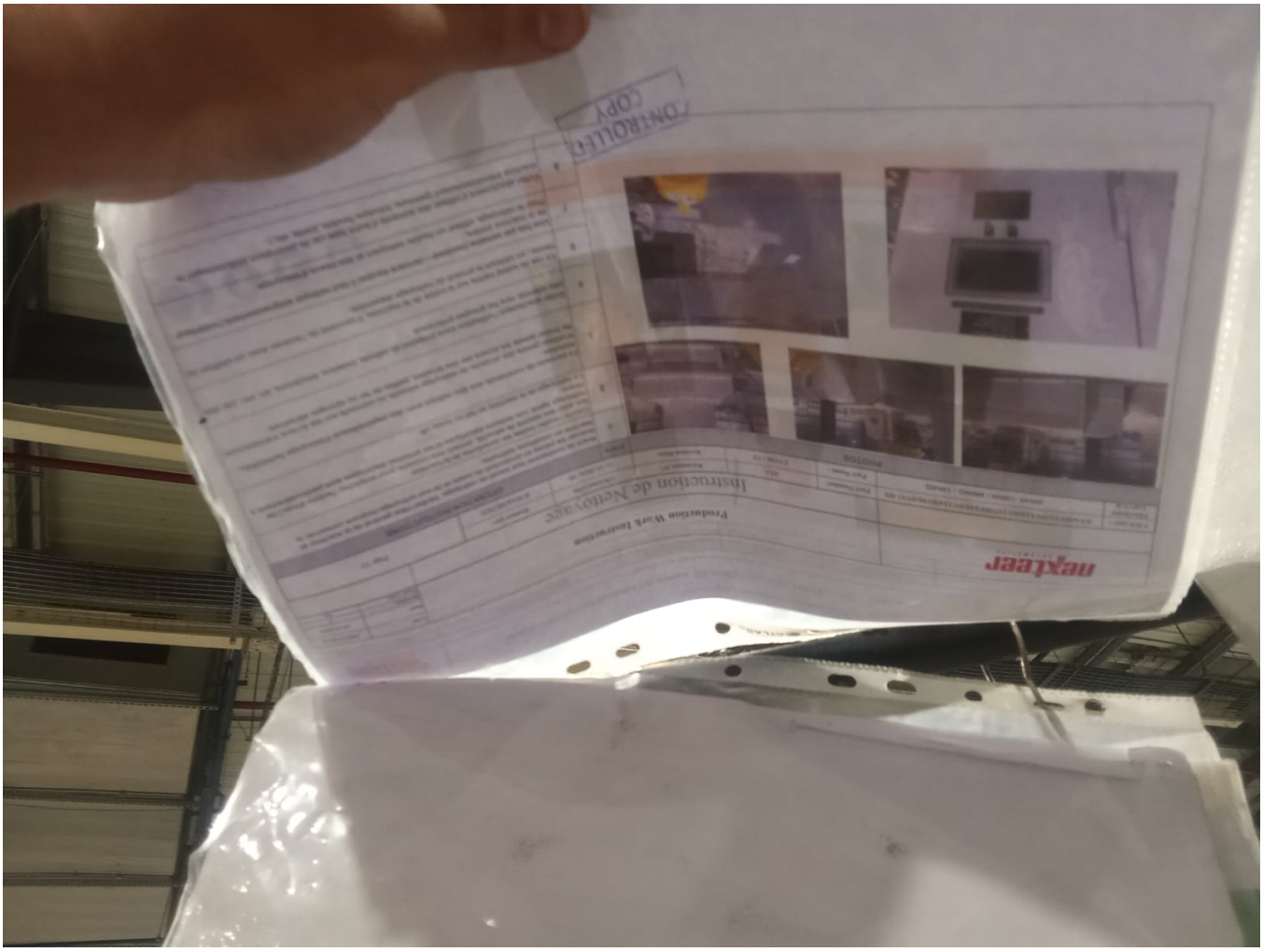
[59/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (5).jpeg**

Content: Instruction de Nettoyage (rotated, in sleeve)



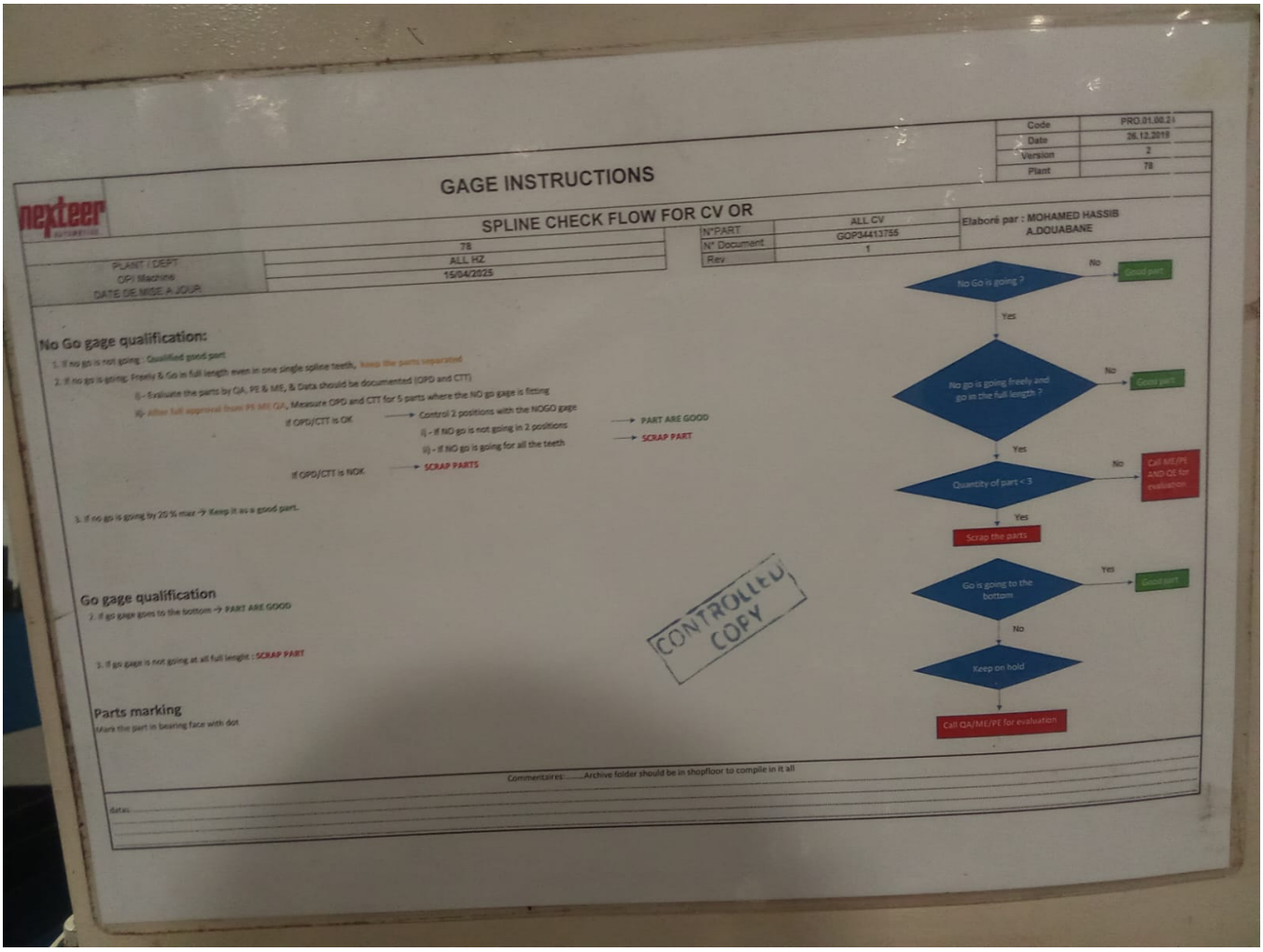
[60/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (6).jpeg**

Content: Gage Instructions — Spline Check Flow for CV OR (GOP34413755) — decision flowchart



[61/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (7).jpeg**

Content: Gage Instructions SG 699436 (GOP34413754) — Max HUB simulator, 4-step check



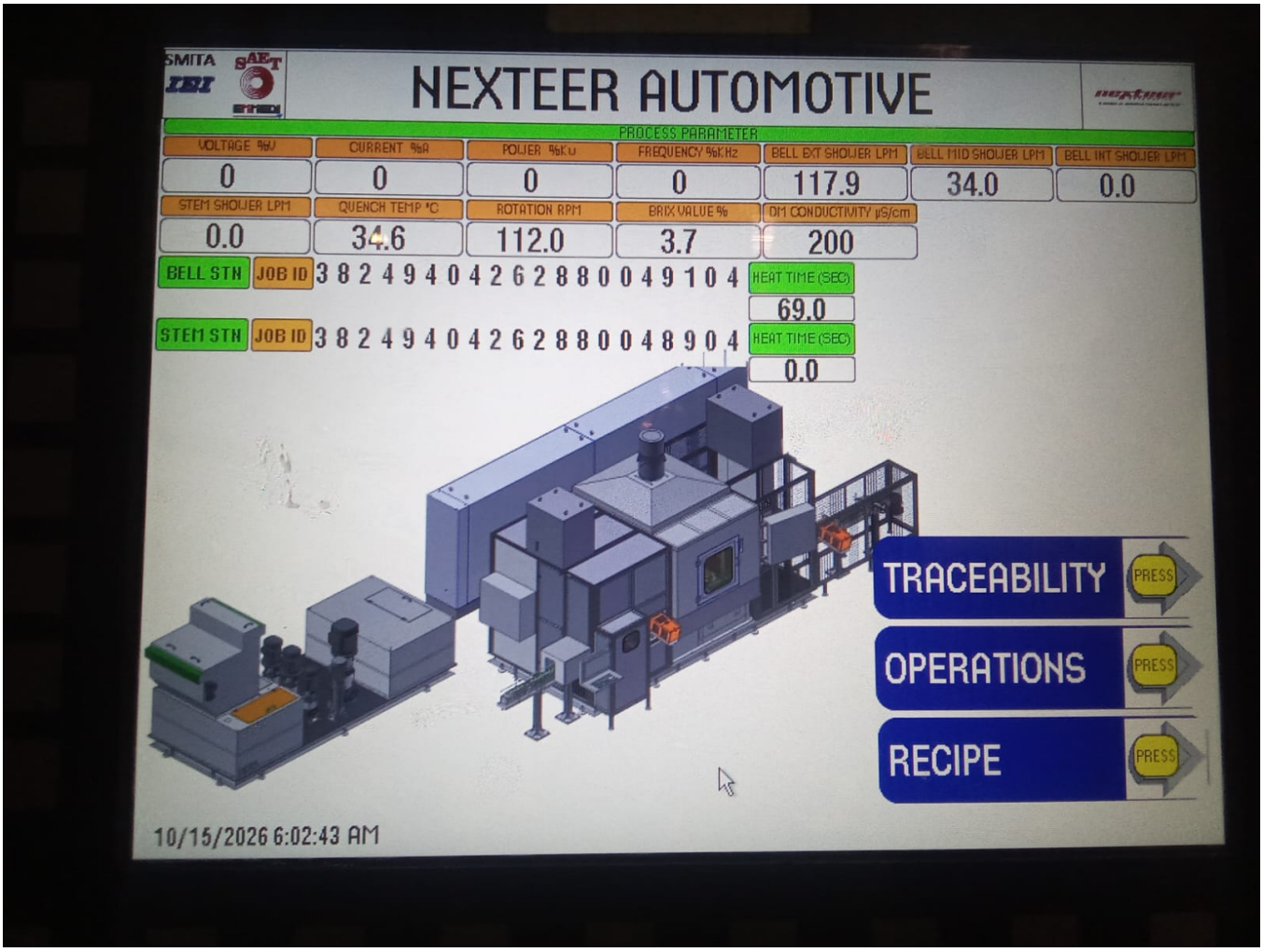
[62/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (8).jpeg**

Content: Inducontrol process dashboard — V/I/P/F, showers, quench 34.6 °C, RPM 112, Brix 3.7 %, conductivity 200, Job IDs, heat time 69 s



[63/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (9).jpeg**

Content: Set Up Sheet (GWI34417348) — draw oven 163 ±5 °C ×3 zones, blower/conveyor Hz



[64/64] **WhatsApp Image 2026-10-15 at 4.21.03 PM (10).jpeg**

Content: Fiche de Sécurité (rotated) — same content

